

Certificate

Issue Date: April 10, 2025

Ref. Report No. ISL-19LE455CE35-MB

Product Name : SATA 2.5"128GB~1TB
Main Model : DS700SSD-128GB
Series Model : DS700SSD-256GB; DS700SSD-512GB; DS700SSD-1TB
Brand : DATO
Responsible Party : DATOTEK INTERNATIONAL Co., Ltd.
Address : No. 34, Alley 18, Lane 478, Rueiguang Rd., Neihs District, Taipei, Taiwan

We, **International Standards Laboratory Corp.**, hereby certify that:

The sample ISL received which bearing the trade name and model specified above has been shown to comply with the applicable technical standards as indicated in the measurement report and was tested in accordance with the measurement procedures specified in European Council Directive EMC Directive 2014/30/EU. And Our laboratories is the accredited laboratories and are approved according to ISO/IEC 17025. The device was passed the test performed according to :



Standards:

EN 55032:2015+AC: 2016, CISPR 32: 2015+COR1:2016: Class B
AS/NZS CISPR 32:2015: Class B
EN 61000-3-2:2014 and IEC 61000-3-2:2014
EN 61000-3-3:2013 and IEC 61000-3-3: 2013
EN 55035: 2017 and CISPR 35: 2016
EN 61000-4-2: 2009 and IEC 61000-4-2: 2008
EN 61000-4-3: 2006+A1: 2008 +A2: 2010 and
IEC 61000-4-3: 2006+A1: 2007+A2: 2010
EN 61000-4-4: 2012 and IEC 61000-4-4:2012
EN 61000-4-5: 2014+A1:2017 and IEC 61000-4-5: 2014+A1:2017
EN 61000-4-6: 2014+AC: 2015 and IEC 61000-4-6:2013
EN 61000-4-8: 2010 and IEC 61000-4-8: 2009
EN 61000-4-11: 2004+A1:2017 and IEC 61000-4-11: 2004+A1:2017

I attest to the accuracy of data and all measurements reported herein were performed by me or were made under my supervision and are correct to the best of my knowledge and belief. I assume full responsibility for the completeness of these measurements and vouch for the qualifications of all persons taking them.

Benson Chen / Sr. Manager

International Standards Laboratory Corp. LT Lab.

TEL: +886-3-263-8888 FAX: +886-3-263-8899

No. 120, Lane 180, Hsin Ho Rd., Lung-Tan Dist., Tao Yuan City 325, Taiwan

CE MARK TECHNICAL FILE

AS/NZS EMC CONSTRUCTION FILE

of

Product Name

SATA 2.5"128GB~1TB

Main Model

DS700SSD-128GB

Series Model

DS700SSD-256GB; DS700SSD-512GB; DS700SSD-1TB

Brand

DATO

Contains:

1. Declaration of Conformity
2. EN 55032/CISPR 32, AS/NZS CISPR 32 EMI test report
3. EN 55035/CISPR 35, EN 61000-3-2 / IEC 61000-3-2, and EN 61000-3-3 / IEC 61000-3-3 test report
4. Block Diagram and Schematics
5. Users' manual

Declaration of Conformity

Name of Responsible Party: DATOTEK INTERNATIONAL Co., Ltd.

Address of Responsible Party: No. 34, Alley 18, Lane 478, Rueiguang Rd.,
Neihu District, Taipei, Taiwan

Declares that product: SATA 2.5"128GB~1TB

Main Model: DS700SSD-128GB

Series Model: DS700SSD-256GB; DS700SSD-512GB;
DS700SSD-1TB

Assembled by: Same as above

Address: Same as above

Conforms to the EMC Directive 2014/30/EU as attested by conformity with the following harmonized standards:

EN 55032:2015+AC:2016, CISPR 32: 2015+COR1:2016: Class B: Electromagnetic compatibility of multimedia equipment - Emission requirements.

AS/NZS CISPR 32:2015: Class B: Electromagnetic compatibility of multimedia equipment- Emission requirements

Performed Item	Test Performed	Deviation	Result
Conducted emissions from the AC mains power ports	Yes	No	PASS
Telecommunication Port Conducted Emissions (asymmetric mode)	N/A	N/A	N/A
Radiated emissions at frequencies below 1 GHz	Yes	No	PASS
Radiated emissions at frequencies above 1 GHz	Yes	No	PASS
Radiated emissions from FM receivers	N/A	N/A	N/A
Voltage Disturbance Emissions at Antenna Terminals	N/A	N/A	N/A
Differential voltage emissions	N/A	N/A	N/A
Outdoor units of home satellite receiving systems	N/A	N/A	N/A

<to be continued>

EN 55035:2017 and CISPR 35:2016: Information technology equipment-Immunity characteristics - Limits and methods of measurement.

Standard	Description	Results	Criteria
EN 61000-4-2:2009 IEC 61000-4-2:2008	Electrostatic Discharge	Pass	B
EN 61000-4-3:2006+A1:2008 +A2:2010 IEC 61000-4-3:2006+A1:2007+A2:2010	Radio-Frequency, Electromagnetic Field	Pass	A
EN 61000-4-4:2012 IEC 61000-4-4:2012	Electrical Fast Transient/Burst	Pass	B
EN 61000-4-5:2014+A1:2017 IEC 61000-4-5:2014+A1:2017	Surge	Pass	B
EN 61000-4-6:2014+AC:2015 IEC 61000-4-6:2013	Conductive Disturbance	Pass	A
EN 61000-4-8:2010 IEC 61000-4-8:2009	Power Frequency Magnetic Field	Pass	A
EN 61000-4-11:2004+A1:2017 IEC 61000-4-11:2004+A1:2017	Voltage Dips / Short Interruption and Voltage Variation		
	>95% in 0.5 cycle	Pass	B
	30% in 25 cycle	Pass	C
	>95% in 250 cycle	Pass	C

Standard	Description	Results
EN 61000-3-2:2014 IEC 61000-3-2:2014	Limits for harmonics current emissions	Pass
EN 61000-3-3:2013 IEC 61000-3-3:2013	Limits for voltage fluctuations and flicker in low-voltage supply systems.	Pass

We, DATOTEK INTERNATIONAL Co., Ltd., hereby declare that the equipment bearing the trade name and model number specified above was tested conforming to the applicable Rules under the most accurate measurement standards possible, and that all the necessary steps have been taken and are in force to assure that production units of the same equipment will continue to comply with the requirements.

DATOTEK INTERNATIONAL Co., Ltd.
Date: April 10, 2025

Remarks: 1) The responsible party for Declaration of Conformity must be located within Europe, 2) The above is a sample of DoC, one should modify it to meet remark 1.

Declaration of Conformity

Name of Responsible Party: DATOTEK INTERNATIONAL Co., Ltd.
Address of Responsible Party: No. 34, Alley 18, Lane 478, Rueiguang Rd.,
Neihu District, Taipei, Taiwan
Declares that product: SATA 2.5"128GB~1TB
Main Model: DS700SSD-128GB
Series Model: DS700SSD-256GB; DS700SSD-512GB;
DS700SSD-1TB
Assembled by: Same as above
Address: Same as above

Conforms to the EMI part of RCM Mark requirements as attested by conformity with the following standards:

AS/NZS CISPR 32:2015: Class B: Electromagnetic compatibility of multimedia equipment- Emission requirements

We, DATOTEK INTERNATIONAL Co., Ltd., hereby declare that the equipment bearing the trade name and model number specified above was tested conforming to the applicable Rules under the most accurate measurement standards possible, and that all the necessary steps have been taken and are in force to assure that production units of the same equipment will continue to comply with the requirements.

DATOTEK INTERNATIONAL Co., Ltd.
Date: April 10, 2025

CE TEST REPORT

of

**EN 55032 / CISPR 32 / AS/NZS CISPR 32
Class B**

**EN 55035 / CISPR 35 / IMMUNITY
EN 61000-3-2 / EN 61000-3-3**

Product: **SATA 2.5"128GB~1TB**
Main Model: **DS700SSD-128GB**
Series Model: **DS700SSD-256GB; DS700SSD-512GB;
DS700SSD-1TB**
Brand: **DATO**
Applicant: **DATOTEK INTERNATIONAL Co., Ltd.**
Address: **No. 34, Alley 18, Lane 478, Rueiguang Rd.,
Neihu District, Taipei, Taiwan**

Test Performed by:



International Standards Laboratory Corp. LT Lab.

TEL: +886-3-263-8888 FAX: +886-3-263-8899

No. 120, Lane 180, Hsin Ho Rd., Lung-Tan Dist., Tao Yuan City 325,
Taiwan

Report No.: **ISL-19LE455CE35-MB**
Issue Date : **April 10, 2025**



Test results given in this report apply only to the specific sample(s) tested and are traceable to national or international standard through calibration of the equipment and evaluating measurement uncertainty herein.
This test report shall not be reproduced except in full, without the written approval of International Standards Laboratory Corp.

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ISL Draft

1. General

1.1 Certification of Accuracy of Test Data

Standards: Please refer to 1.2

Equipment Tested: SATA 2.5"128GB~1TB

Main Model: DS700SSD-128GB

Series Model: DS700SSD-256GB; DS700SSD-512GB; DS700SSD-1TB

Brand: DATO

Applicant: DATOTEK INTERNATIONAL Co., Ltd.

Sample received Date: August 2, 2019

Final test Date: EMI: refer to the date of test data
EMS: August 13, 2019

Test Site: Chamber 02; Chamber 14; Conduction 03; Immunity 02

Test Distance: 10M; 3M (above1GHz) (EMI test)

Temperature: refer to each site test data

Humidity: refer to each site test data

Atmospheric Pressure: 86 kPa to 106 kPa

Input power: Conduction input power: AC 230 V / 50 Hz
Radiation input power: AC 230 V / 50 Hz
Immunity input power: AC 230 V / 50 Hz

Test Result: PASS

Report Engineer: Jayla Lu

Test Engineer: Jovi Liu
Jovi Liu

Approved By: Angus Chu
Angus Chu / Sr. Manager

1.2 Test Standards

The tests which this report describes were conducted by an independent electromagnetic compatibility consultant, International Standards Laboratory Corp. in accordance with the following

EN 55032:2015+AC:2016, CISPR 32: 2015+COR1:2016: Class B: Electromagnetic compatibility of multimedia equipment - Emission requirements.

AS/NZS CISPR 32:2015: Class B: Electromagnetic compatibility of multimedia equipment- Emission requirements

Performed Item	Test Performed	Deviation	Result
Conducted emissions from the AC mains power ports	Yes	No	PASS
Telecommunication Port Conducted Emissions (asymmetric mode)	N/A	N/A	N/A
Radiated emissions at frequencies below 1 GHz	Yes	No	PASS
Radiated emissions at frequencies above 1 GHz	Yes	No	PASS
Radiated emissions from FM receivers	N/A	N/A	N/A
Voltage Disturbance Emissions at Antenna Terminals	N/A	N/A	N/A
Differential voltage emissions	N/A	N/A	N/A
Outdoor units of home satellite receiving systems	N/A	N/A	N/A

EN 55035:2017 and CISPR 35:2016: Information technology equipment-Immunity characteristics - Limits and methods of measurement.

Standard	Description	Results	Criteria
EN 61000-4-2:2009 IEC 61000-4-2:2008	Electrostatic Discharge	Pass	B
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EN 61000-4-4:2012 IEC 61000-4-4:2012	Electrical Fast Transient/Burst	Pass	B
EN 61000-4-5:2014+A1:2017 IEC 61000-4-5:2014+A1:2017	Surge	Pass	B
EN 61000-4-6:2014+AC:2015 IEC 61000-4-6:2013	Conductive Disturbance	Pass	A
EN 61000-4-8:2010 IEC 61000-4-8:2009	Power Frequency Magnetic Field	Pass	A
EN 61000-4-11:2004+A1:2017 IEC 61000-4-11:2004+A1:2017	Voltage Dips / Short Interruption and Voltage Variation		
	>95% in 0.5 cycle	Pass	B
	30% in 25 cycle	Pass	C
	>95% in 250 cycle	Pass	C
Standard	Description	Results	
EN 61000-3-2:2014 IEC 61000-3-2:2014	Limits for harmonics current emissions	Pass	
EN 61000-3-3:2013 IEC 61000-3-3:2013	Limits for voltage fluctuations and flicker in low-voltage supply systems.	Pass	

1.2.1 Performance Criteria for Compliance: EN 55035

Performance criterion A

The equipment shall continue to operate as intended without operator intervention. No degradation of performance, loss of function or change of operating state is allowed below a performance level specified by the manufacturer when the equipment is used as intended. The performance level may be replaced by a permissible loss of performance. If the minimum performance level or the permissible performance loss is not specified by the manufacturer, then either of these may be derived from the product description and documentation, and by what the user may reasonably expect from the equipment if used as intended.

Performance criterion B

During the application of the disturbance, degradation of performance is allowed. However, no unintended change of actual operating state or stored data is allowed to persist after the test.

After the test, the equipment shall continue to operate as intended without operator intervention; no degradation of performance or loss of function is allowed, below a performance level specified by the manufacturer, when the equipment is used as intended. The performance level may be replaced by a permissible loss of performance.

If the minimum performance level (or the permissible performance loss), or recovery time, is not specified by the manufacturer, then either of these may be derived from the product description and documentation, and by what the user may reasonably expect from the equipment if used as intended.

Performance criterion C

Loss of function is allowed, provided the function is self-recoverable, or can be restored by the operation of the controls by the user in accordance with the manufacturer's instructions. A reboot or re-start operation is allowed.

Information stored in non-volatile memory, or protected by a battery backup, shall not be lost.

1.3 Description of EUT

EUT

This report test data using the report number ISL-19LE455CE35

Description	SATA 2.5"128GB~1TB
Condition	Pre-Production
Main Model	DS700SSD-128GB
Series Model	DS700SSD-256GB; DS700SSD-512GB; DS700SSD-1TB
Serial Number	N/A
Power	From Personal Computer Supply
Control IC	PS3111-S11
PCB Number	A117011NP000033N-V3
Memory Capacity	1TB
Highest working frequency	6GHz
The radiation test should be tested till 6GHz	

The memory capacities listed below is chosen by the applicant to be the representative configuration for testing in this report.

Test configuration:

Configuration	Control IC	PCB Number	Memory Capacity
1	PS3111-S11	A117011NP000033N-V3	1TB

Different Model list:

Model	Market
DS700SSD-128GB; DS700SSD-256GB; DS700SSD-512GB; DS700SSD-1TB	Different customer

EMI Noise Source:

Refer to the photo	Crystal	Point
EUT-4	25MHz	Y1

EMI Solution:

NA

1.4 Description of Support Equipment

For EMI test Configuration Support unit: 1~6

For EMS test Configuration Support unit: 2~7

No	Unit	Model / Serial No.	Brand	Power Cord	FCC ID
1	LCD Monitor	P2416D S/N: N/A	DELL	Non-shielded	FCC DOC
2	PS/2 Keyboard	Y-S0002 S/N: N/A	Logitech	N/A	FCC DOC
3	PS/2 Mouse	M-SBM96B S/N: N/A	Logitech	N/A	FCC DOC
4	Printer	C930 S/N: N/A	HP	Non-shielded	FCC DOC
5	Modem	DM1414 S/N: N/A	Aceex	Non-shielded	FCC DOC
6	Personal Computer	HP ProDesk 400 G3 MT S/N: N/A	HP	Non-shielded	FCC DOC
7	LCD Monitor	ST2420L S/N: N/A	DELL	Non-shielded	FCC DOC

1.5 Software for Controlling Support Unit

Test programs exercising various part of EUT were used. The programs were executed as follows:

1. PC running Winthrax.exe to read and write the EUT.
2. Send signal to the Printer through PC USB Port.
3. Send signal to the Modem through PC Serial Port.
4. Send Color Bar to the LCD Monitor through PC D-SUB Port.
5. Repeat the above steps.

	Filename	Issued Date
EUT	Winthrax.exe	06/14/2005
LCD Monitor	Media Player.exe	11/04/2009
Printer	IntelEMC	04/11/2007
Modem	IntelEMC	04/11/2007

1.6 I/O Cable Condition of EUT and Support Units

Description	Path	Cable Length	Cable Type
AC Power Cable	100V (~240V) to PC SPS	1.8m	Non-shielded
Monitor Data Cable	Monitor D-SUB Port to PC D-SUB Port	1.8m	Shielded (with core)
Keyboard Data Cable	PS/2 Keyboard to PC PS/2 Port	1.8m	Shielded
Mouse Data Cable	PS/2 Mouse to PC PS/2 Port	1.8m	Shielded
Printer Data Cable	Printer to PC USB Port	1.8m	Shielded
Modem Data Cable	Modem to PC Serial Port	1.8m	Shielded

2.1.4 Limit

Conducted emissions from the AC mains power ports of Class_A equipment:

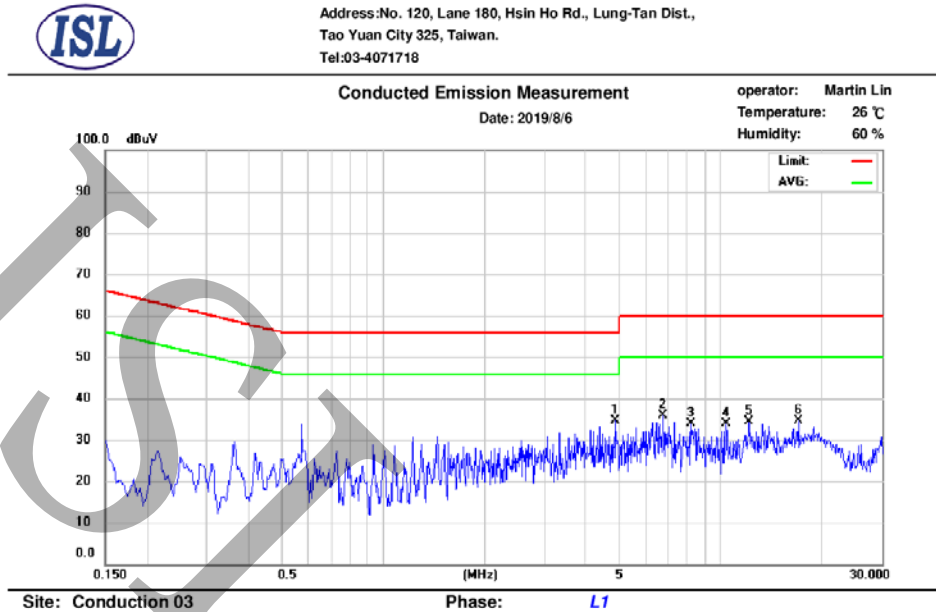
Frequency	QP	AV
MHz	dB(μ V)	dB(μ V)
0.15-0.50	79	66
5.0-30	73	60
Note: The lower limit shall apply at the transition frequencies		

Conducted emissions from the AC mains power ports of Class_B equipment:

Frequency	QP	AV
MHz	dB(μ V)	dB(μ V)
0.15-0.50	66-56	56-46
0.50-5.0	56	46
5.0-30	60	50
Note: The lower limit shall apply at the transition frequencies		

2.2 Conduction Test Data: Configuration 1

-Live



No.	Frequency (MHz)	QP_R (dBuV)	AVG_R (dBuV)	Correct Factor (dB)	QP Emission (dBuV)	QP Limit (dBuV)	QP Margin (dB)	AVG Emission (dBuV)	AVG Limit (dBuV)	AVG Margin (dB)
1	4.898	20.08	12.81	9.74	29.82	56.00	-26.18	22.55	46.00	-23.45
2	6.742	22.99	17.90	9.77	32.76	60.00	-27.24	27.67	50.00	-22.33
3	8.166	21.46	17.20	9.80	31.26	60.00	-28.74	27.00	50.00	-23.00
4	10.362	18.71	11.05	9.84	28.55	60.00	-31.45	20.89	50.00	-29.11
5	12.138	20.84	16.01	9.87	30.71	60.00	-29.29	25.88	50.00	-24.12
6	16.970	18.27	13.42	9.93	28.20	60.00	-31.80	23.35	50.00	-26.65

Note:

Margin = QP/AVG Emission - Limit

QP/AVG Emission = QP_R/AVG_R + Correct Factor

Correct Factor = LISN Loss + Cable Loss

A margin of -8dB means that the emission is 8dB below the limit

The frequency spectrum graph is for final peak graph, and the attached table is for QP/AVG test result.

If peak data can pass, it will be shown in "QP/AVG Correct" column, if not, QP/AVG data will instead.

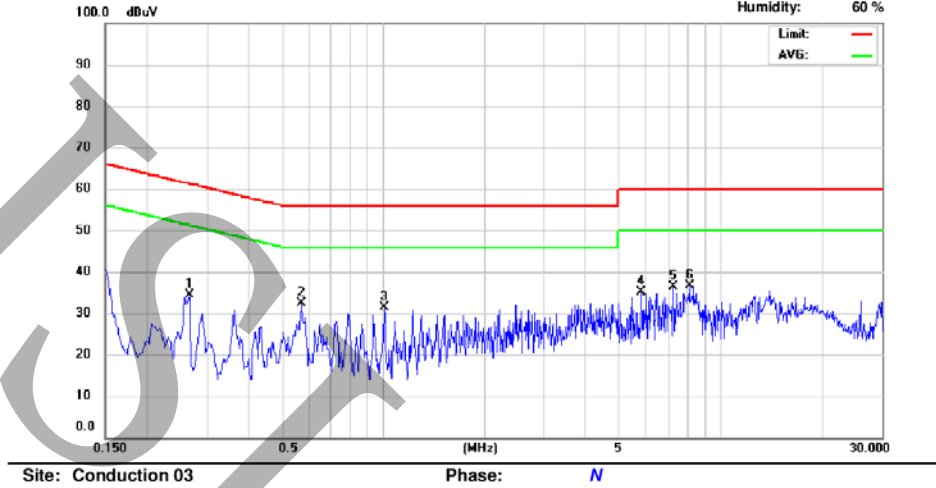
- Neutral



Address: No. 120, Lane 180, Hsin Ho Rd., Lung-Tan Dist.,
Tao Yuan City 325, Taiwan.
Tel: 03-4071718

Conducted Emission Measurement
Date: 2019/8/6

operator: Martin Lin
Temperature: 26 °C
Humidity: 60 %



No.	Frequency (MHz)	QP_R (dBuV)	AVG_R (dBuV)	Correct Factor (dB)	QP Emission (dBuV)	QP Limit (dBuV)	QP Margin (dB)	AVG Emission (dBuV)	AVG Limit (dBuV)	AVG Margin (dB)
1	0.266	19.59	3.88	9.64	29.23	61.24	-32.01	13.52	51.24	-37.72
2	0.574	20.77	14.88	9.65	30.42	56.00	-25.58	24.53	46.00	-21.47
3	1.010	18.02	15.05	9.67	27.69	56.00	-28.31	24.72	46.00	-21.28
4	5.822	20.42	13.39	9.79	30.21	60.00	-29.79	23.18	50.00	-26.82
5	7.242	21.83	15.73	9.82	31.65	60.00	-28.35	25.55	50.00	-24.45
6	8.090	20.30	16.13	9.84	30.14	60.00	-29.86	25.97	50.00	-24.03

Note:

Margin = QP/AVG Emission - Limit

QP/AVG Emission = QP_R/AVG_R + Correct Factor

Correct Factor = LISN Loss + Cable Loss

A margin of -8dB means that the emission is 8dB below the limit

The frequency spectrum graph is for final peak graph, and the attached table is for QP/AVG test result.

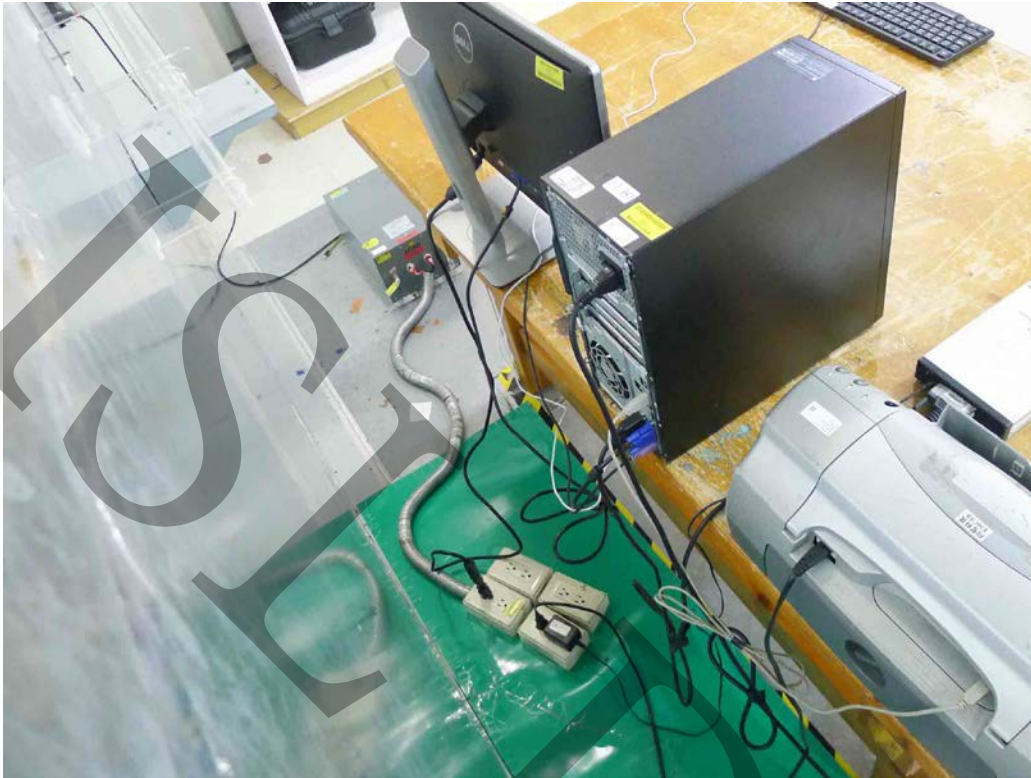
If peak data can pass, it will be shown in "QP/AVG Correct" column, if not, QP/AVG data will instead.

2.3 Test Setup Photo

Front View



Back View



3.1.4 Limit

Asymmetric mode conducted emissions from Class_A equipment:

Applicable to

1. wired network ports.
2. optical fibre ports with metallic shield or tension members.
3. antenna ports.

Frequency range MHz	Coupling device	Detector type / bandwidth	Class_A voltage limits dB(μV)	Class_A current limits dB(μA)
0.15-0.5	AAN	Quasi Peak / 9 kHz	97-87	n/a
0.5-30			87	
0.15-0.5	AAN	Average / 9 kHz	84-74	
0.5-30			74	
0.15-0.5	CVP and current probe	Quasi Peak / 9 kHz	97-87	53-43
0.5-30			87	43
0.15-0.5	CVP and current probe	Average / 9 kHz	84-74	40-30
0.5-30			74	30
0.15-0.5	Current Probe	Quasi Peak / 9 kHz	n/a	53-43
0.5-30				43
0.15-0.5	Current Probe	Average / 9 kHz		40-30
0.5-30				30

Asymmetric mode conducted emissions from Class_B equipment:

Applicable to:

1. wired network ports.
2. optical fibre ports with metallic shield or tension members.
3. broadcast receiver tuner ports.
4. antenna ports.

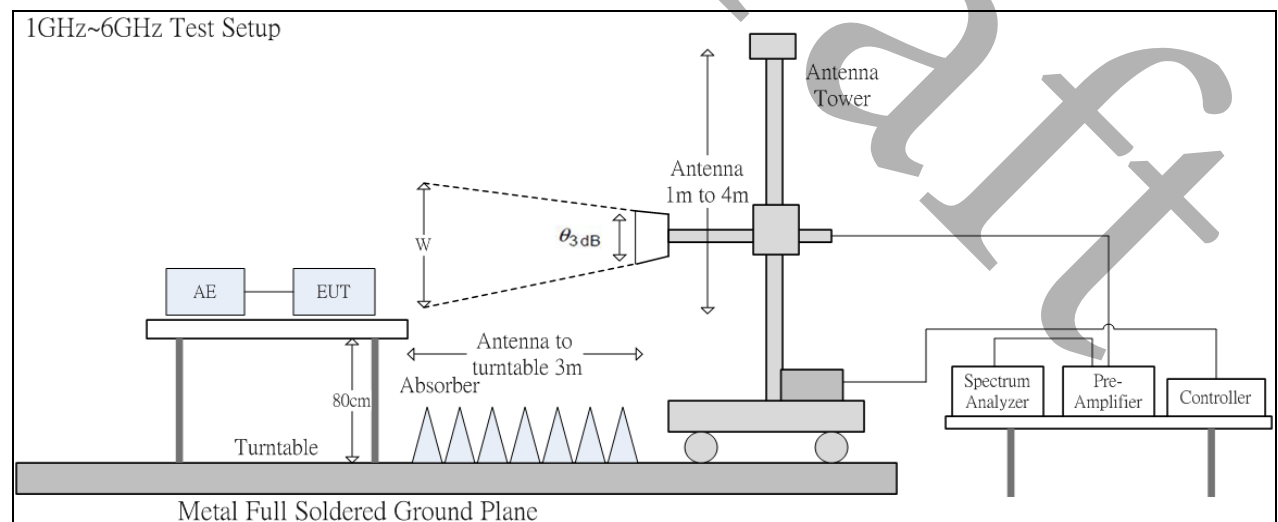
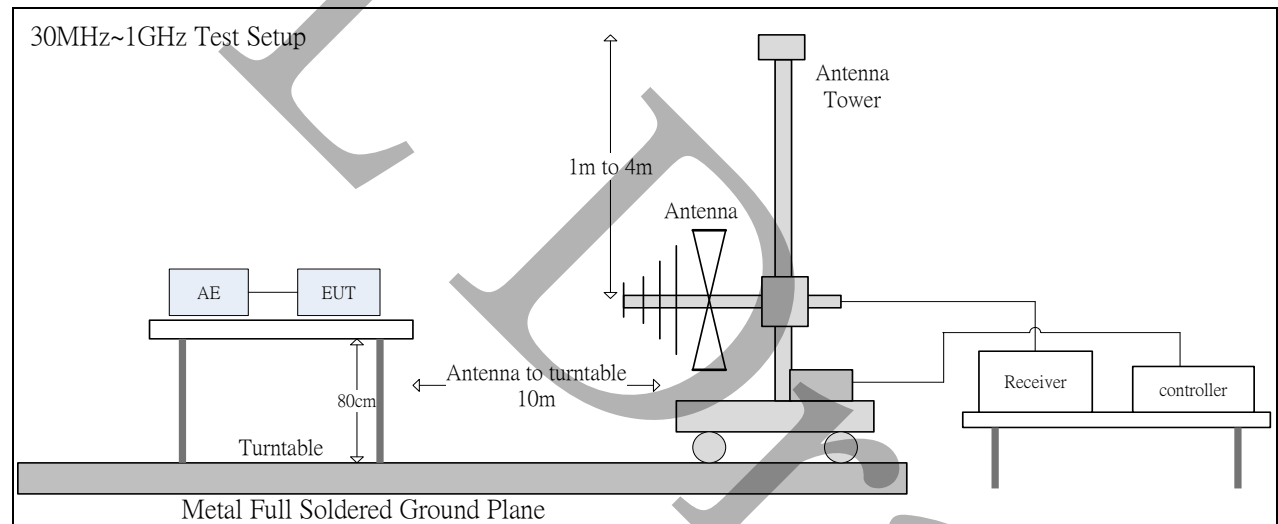
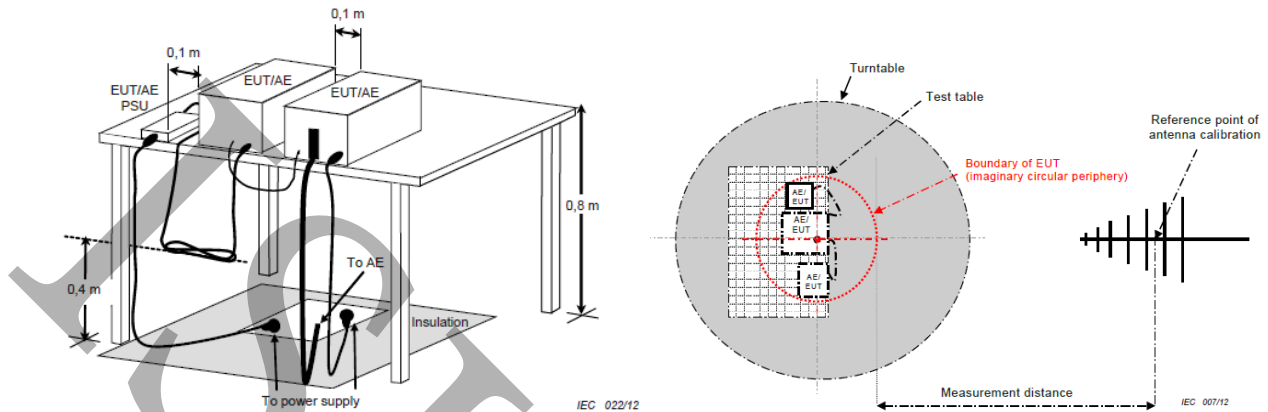
Frequency range MHz	Coupling device	Detector type / bandwidth	Class_B voltage limits dB(μV)	Class_B current limits dB(μA)
0.15-0.5	AAN	Quasi Peak / 9 kHz	84-74	n/a
0.5-30			74	
0.15-0.5	AAN	Average / 9 kHz	74-64	
0.5-30			64	
0.15-0.5	CVP and current probe	Quasi Peak / 9 kHz	84-74	40-30
0.5-30			74	30
0.15-0.5	CVP and current probe	Average / 9 kHz	74-64	30-20
0.5-30			64	20
0.15-0.5	Current Probe	Quasi Peak / 9 kHz	n/a	40-30
0.5-30				30
0.15-0.5	Current Probe	Average / 9 kHz		30-20
0.5-30				20

****Remarks: It is not necessary to be tested on this item.**

4. Radiated Disturbance Emissions

4.1 Test Setup and Procedure

4.1.1 Test Setup



The 3dB beam width of the horn antenna used for the test is as shown in the table below.

Frequency (GHz)	E-plane	H-plane	$\theta_{3dB}(\text{min})$	d= 3 m
				w (m)
1	88°	147°	88°	5.79
2	68°	119°	68°	4.04
3	73°	92°	73°	4.44
4	70°	89°	70°	4.20
5	55°	60°	55°	3.12
6	63°	62°	62°	3.60

4.1.2 Test Procedure

The radiated emissions test will then be repeated on the open site or chamber to measure the amplitudes accurately and without the multiple reflections existing in the shielded room. The EUT and support equipment are set up on the turntable of one of 10 meter open field sites or 10 meter chamber. Desktop EUT are set up on a FRP stand 0.8 meter above the ground or floor-standing arrangement shall be placed on the horizontal ground reference plane.

For the initial measurements, the receiving antenna is varied from 1-4 meter height and is changed in the vertical plane from vertical to horizontal polarization at each frequency. The highest emissions between 30 MHz to 1000 MHz were analyzed in details by operating the spectrum analyzer and/or EMI receiver in quasi-peak mode to determine the precise amplitude of the emissions. The highest emissions between 1 GHz to 6 GHz were analyzed in details by operating the spectrum analyzer in peak and average mode to determine the precise amplitude of the emissions. The test volume for a height of up to 30 cm may be obstructed by absorber placed on the ground plane.

At the highest amplitudes observed, the EUT is rotated in the horizontal plane while changing the antenna polarization in the vertical plane to maximize the reading. The interconnecting cables were arranged and moved to get the maximum measurement. Once the maximum reading is obtained, the antenna elevation and polarization will be varied between specified limits to maximize the readings. All of the interface cables were manipulated according to EN 55032 requirements.

The highest internal source of an EUT is defined as the highest frequency generated or used within the EUT or on which the EUT operates or tunes.

If the highest frequency of the internal sources of the EUT is less than 108 MHz, the measurement shall only be made up to 1 GHz.

If the highest frequency of the internal sources of the EUT is between 108 MHz and 500 MHz, the measurement shall only be made up to 2 GHz.

If the highest frequency of the internal sources of the EUT is between 500 MHz and 1 GHz, the measurement shall only be made up to 5 GHz.

If the highest frequency of the internal sources of the EUT is above 1 GHz, the measurement shall be made up to 5 times the highest frequency or 6 GHz, whichever is less.

4.1.3 Spectrum Analyzer Configuration (for the frequencies tested)

Frequency Range: 30MHz--1000MHz
 Detector Function: Quasi-Peak Mode
 Resolution Bandwidth: 120kHz

Frequency Range: Above 1 GHz to 6 GHz
 Detector Function: Peak/Average Mode
 Resolution Bandwidth: 1MHz

4.2 Limit

Radiated emissions at frequencies up to 1 GHz for Class_A equipment:

Frequency range MHz	Measurement		Class_A limits dB(μV/m)
	Distance m	Detector type / bandwidth	OATS/SAC
30-230	10	Quasi Peak / 120 kHz	40
230-1000			47
30-230	3		50
230-1000			57

Radiated emissions at frequencies above 1 GHz for Class_A equipment:

Frequency range MHz	Measurement		Class_A limits dB(μ V/m)
	Distance m	Detector type / bandwidth	FSOATS
1000-3000	3	Average / 1MHz	56
3000-6000			60
1000-3000		Peak / 1MHz	76
3000-6000			80

Radiated emissions at frequencies up to 1 GHz for Class_B equipment:

Frequency range MHz	Measurement		Class_B limits dB(μV/m)
	Distance m	Detector type / bandwidth	OATS/SAC
30-230	10	Quasi Peak / 120 kHz	30
230-1000			37
30-230	3		40
230-1000			47

Radiated emissions at frequencies above 1 GHz for Class_B equipment:

Frequency range MHz	Measurement		Class_B limits dB(μ V/m)
	Distance m	Detector type / bandwidth	FSOATS
1000-3000	3	Average / 1MHz	50
3000-6000			54
1000-3000		Peak / 1MHz	70
3000-6000			74

Radiated emissions from FM receivers:

Frequency range MHz	Measurement		Class_B limits dB(μV/m)	
	Distance m	Detector type / bandwidth	Fundamental	Harmonics
			OATS/SAC	OATS/SAC
30-230	10	Quasi Peak / 120 kHz	50	42
230-300				42
300-1000				46
30-230	3		60	52
230-300				52
300-1000				56

4.3 Radiation Test Data: Configuration 1 - Radiated Emissions (Horizontal)



Address: No. 120, Lane 180, Hsin Ho Rd., Lung-Tan Dist.,
Tao Yuan City 325, Taiwan.
Tel: 03-4071718

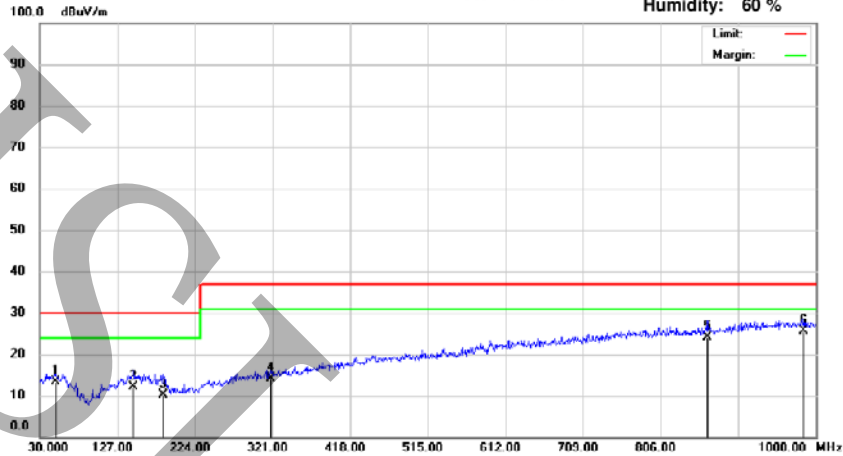
Radiated Emission Measurement

Date: 2019/8/8

Operator: Elric Chen

Temperature: 26 °C

Humidity: 60 %



Site : Chamber 02

Polarization: Horizontal

Mk.	Frequency (MHz)	RX_R (dBuV)	Correct Factor (dB/m)	Emission (dBuV/m)	Limit (dBuV/m)	Margin (dB)	Ant.Pos (cm)	Tab.Pos (deg.)	Detector
1	49.40	30.21	-16.92	13.29	30.00	-16.71	399	360	QP
2	147.37	28.14	-16.13	12.01	30.00	-17.99	400	6	QP
3	184.23	27.91	-17.75	10.16	30.00	-19.84	270	0	QP
4	319.06	28.32	-14.19	14.13	37.00	-22.87	300	53	QP
5	864.20	27.91	-3.82	24.09	37.00	-12.91	400	283	QP
6	985.45	27.47	-1.95	25.52	37.00	-11.48	200	152	QP

* Note:

Margin = Emission – Limit

Emission = Radiated Amplitude + Correct Factor

Correct Factor = Antenna Correction Factor + Cable Loss – Pre-Amplifier Gain

A margin of -8dB means that the emission is 8dB below the limit

Antenna Distance: 10 meters

Below 1GHz test, if the peak measured value meets the QP limit, it is unnecessary to perform the QP measurement.



Address: No. 120, Lane 180, Hsin Ho Rd., Lung-Tan Dist.,
Tao Yuan City 325, Taiwan.
Tel: 03-4071718

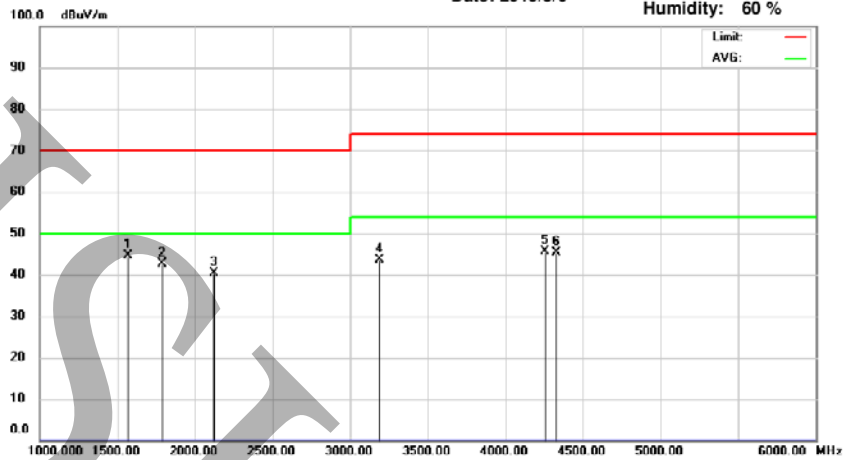
Radiated Emission Measurement

Date: 2019/8/6

Operator: Jerry Su

Temperature: 26 °C

Humidity: 60 %



Site : Chamber 14

Polarization: *Horizontal*

Mk.	Frequency (MHz)	RX_R (dBuV)	Correct Factor(dB/m)	Emission (dBuV/m)	Limit (dBuV/m)	Margin (dB)	Ant.Pos (cm)	Tab.Pos (deg.)	Detector
1	1570.00	60.60	-16.05	44.55	70.00	-25.45	138	234	peak
2	1790.00	56.59	-14.01	42.58	70.00	-27.42	288	206	peak
3	2125.00	53.23	-12.75	40.48	70.00	-29.52	304	227	peak
4	3190.00	54.40	-10.65	43.75	74.00	-30.25	394	119	peak
5	4255.00	55.82	-10.15	45.67	74.00	-28.33	100	106	peak
6	4330.00	55.54	-10.28	45.26	74.00	-28.74	157	197	peak

* Note:

Margin = Emission – Limit

Emission = Radiated Amplitude + Correct Factor

Correct Factor = Antenna Correction Factor + Cable Loss – Pre-Amplifier Gain

A margin of -8dB means that the emission is 8dB below the limit

Antenna Distance: 3 meters

Above 1GHz test, if the peak measured value meets the average limit, it is unnecessary to perform the average measurement.

-Radiated Emissions (Vertical)



Address: No. 120, Lane 180, Hsin Ho Rd., Lung-Tan Dist.,
Tao Yuan City 325, Taiwan.
Tel: 03-4071718

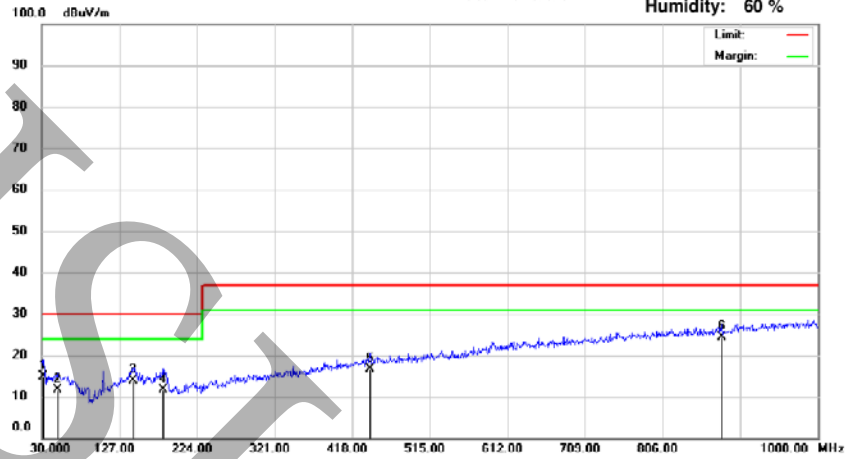
Radiated Emission Measurement

Date: 2019/8/8

Operator: Elric Chen

Temperature: 26 °C

Humidity: 60 %



Site : Chamber 02

Polarization: Vertical

Mk.	Frequency (MHz)	RX_R (dBuV)	Correct Factor (dB/m)	Emission (dBuV/m)	Limit (dBuV/m)	Margin (dB)	Ant.Pos (cm)	Tab.Pos (deg.)	Detector
1	31.94	33.68	-18.87	14.81	30.00	-15.19	100	53	QP
2	49.40	28.48	-16.92	11.56	30.00	-18.44	200	357	QP
3	144.46	30.31	-16.32	13.99	30.00	-16.01	100	360	QP
4	181.32	28.91	-17.40	11.51	30.00	-18.49	100	0	QP
5	440.31	27.52	-10.96	16.56	37.00	-20.44	100	251	QP
6	879.72	27.99	-3.57	24.42	37.00	-12.58	200	55	QP

* Note:

Margin = Emission – Limit

Emission = Radiated Amplitude + Correct Factor

Correct Factor = Antenna Correction Factor + Cable Loss – Pre-Amplifier Gain

A margin of -8dB means that the emission is 8dB below the limit

Antenna Distance: 10 meters

Below 1GHz test, if the peak measured value meets the QP limit, it is unnecessary to perform the QP measurement.

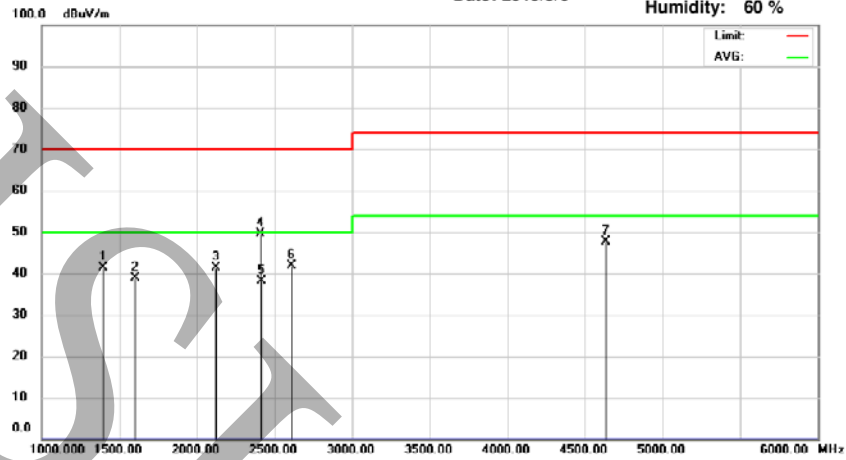


Address: No. 120, Lane 180, Hsin Ho Rd., Lung-Tan Dist.,
Tao Yuan City 325, Taiwan.
Tel: 03-4071718

Radiated Emission Measurement

Date: 2019/8/6

Operator: Jerry Su
Temperature: 26 °C
Humidity: 60 %



Site : Chamber 14

Polarization: Vertical

Mk.	Frequency (MHz)	RX_R (dBuV)	Correct Factor(dB/m)	Emission (dBuV/m)	Limit (dBuV/m)	Margin (dB)	Ant.Pos (cm)	Tab.Pos (deg.)	Detector
1	1395.00	57.13	-15.75	41.38	70.00	-28.62	100	162	peak
2	1600.00	54.69	-15.86	38.83	70.00	-31.17	172	0	peak
3	2125.00	54.22	-12.75	41.47	70.00	-28.53	201	188	peak
4	2410.00	61.64	-12.01	49.63	70.00	-20.37	178	0	peak
5	2413.10	50.17	-12.00	38.17	50.00	-11.83	150	22	AVG
6	2610.00	53.21	-11.39	41.82	70.00	-28.18	201	355	peak
7	4635.00	57.84	-10.23	47.61	74.00	-26.39	151	182	peak

* Note:

Margin = Emission – Limit

Emission = Radiated Amplitude + Correct Factor

Correct Factor = Antenna Correction Factor + Cable Loss – Pre-Amplifier Gain

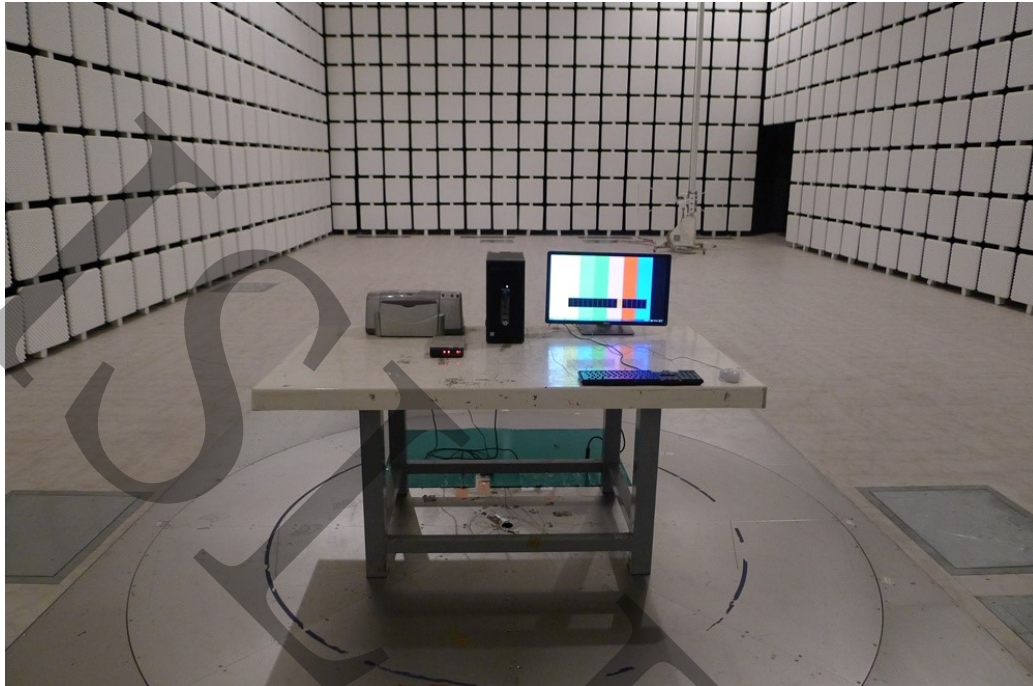
A margin of -8dB means that the emission is 8dB below the limit

Antenna Distance: 3 meters

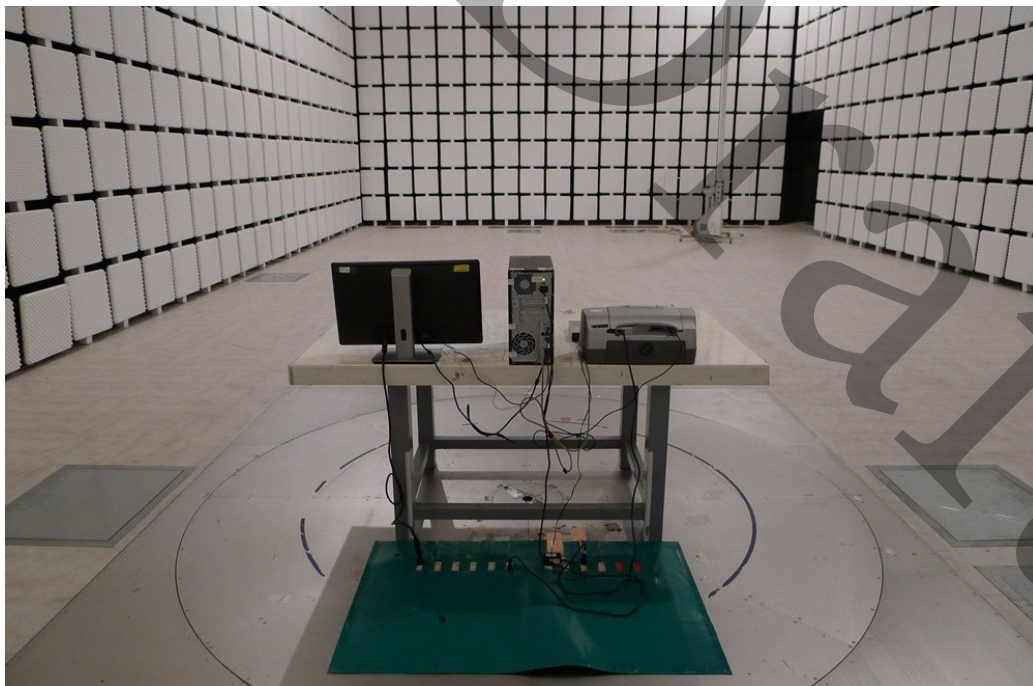
Above 1GHz test, if the peak measured value meets the average limit, it is unnecessary to perform the average measurement.

4.4 Test Setup Photo

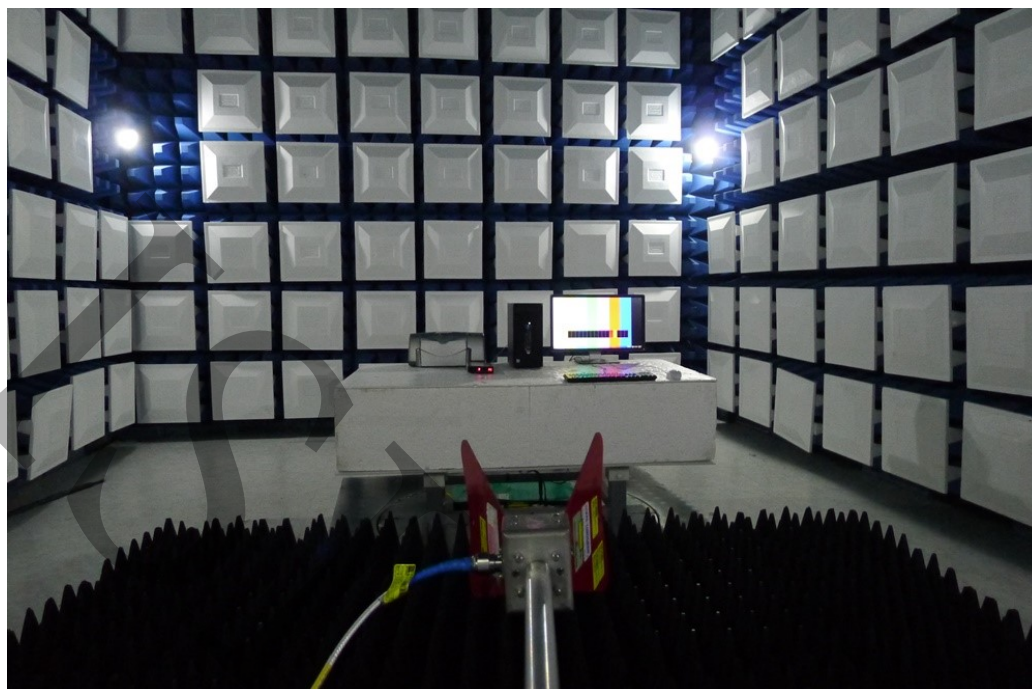
Front View (30MHz~1GHz)



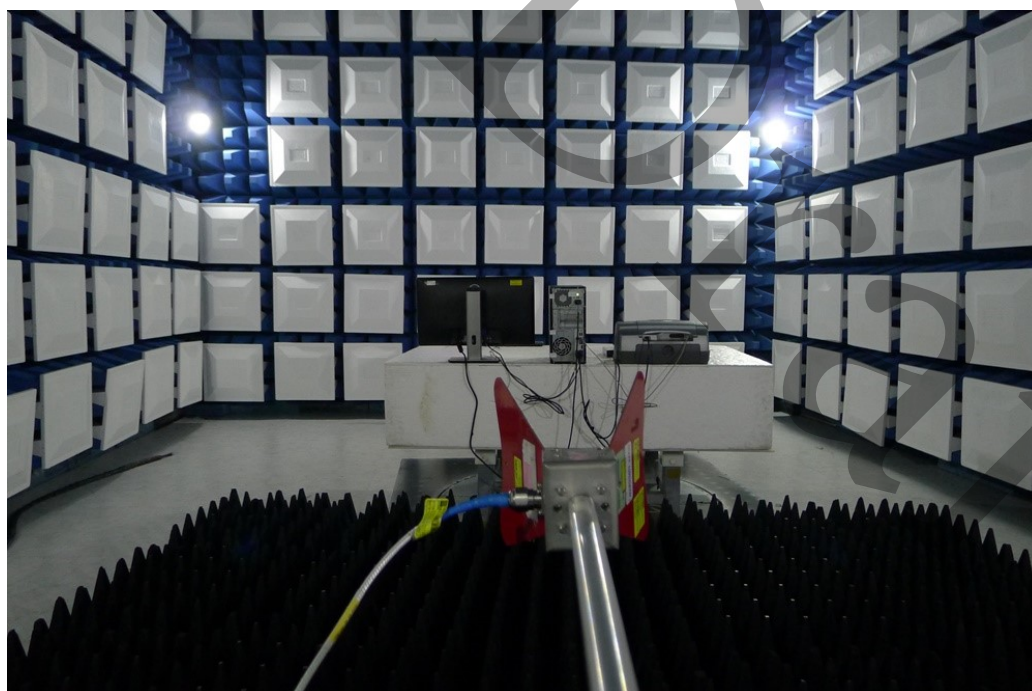
Back View (30MHz~1GHz)



Front View (above 1GHz)



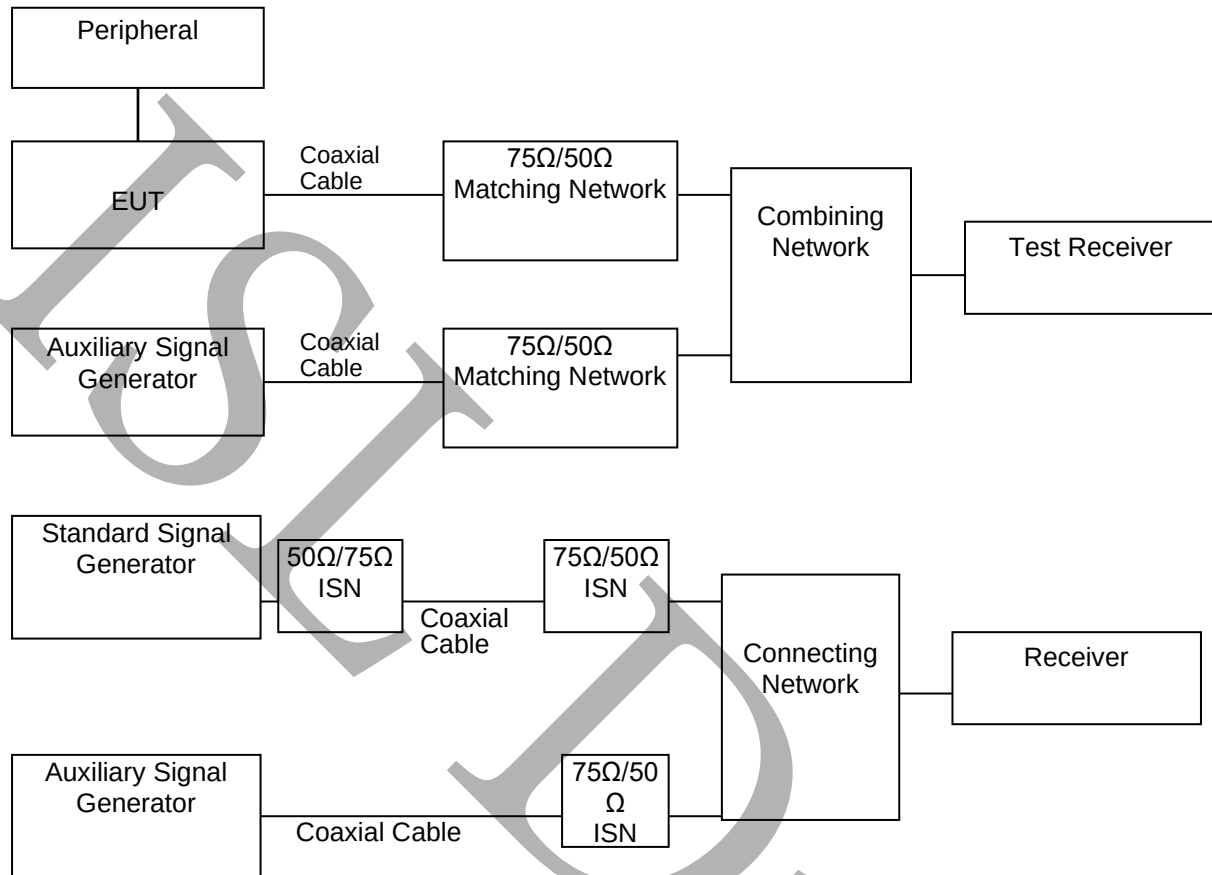
Back View (above 1GHz)



5. Voltage Disturbance Emissions at Antenna Terminals

5.1 Test Setup and Procedure

5.1.1 Test Setup



5.1.2 Test Procedure

The output level of the auxiliary signal generator was set to 70dBuV at the EUT antenna terminal with 75 ohms impedance with an un-modulated carrier.

The highest emissions were analyzed in details by operating the spectrum analyzer and/or EMI receiver in quasi-peak mode to determine the precise amplitude of the emissions. The power of EUT was switched off to make sure the emission was not contributed by the auxiliary signal generator. While doing so, the interconnecting cables and major parts of the system were moved around to maximize the emission.

5.1.3 EMI Receiver Configuration (for the frequencies tested)

Frequency Range:	30MHz-2150MHz
Detector Function:	Quasi-Peak Mode
Resolution Bandwidth:	120kHz

5.1.4 Limit

Applicable to:

1. TV broadcast receiver tuner ports with an accessible connector.
2. RF modulator output ports.
3. FM broadcast receiver tuner ports with an accessible connector.

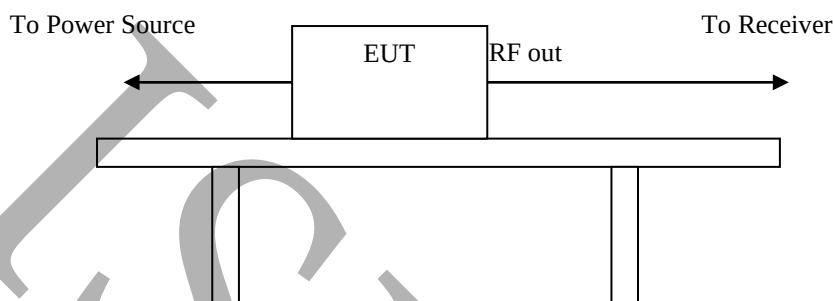
Table clause	Frequency range MHz	Detector type/ bandwidth	Class B limits dB(μV) 75 Ω			Applicability
			Other	Local Oscillator Fundamental	Local Oscillator Harmonics	
A12.1	30 – 950	For frequencies ≤1 GHz	46	46	46	See a)
	950 – 2 150		46	54	54	
A12.2	950 – 2 150	Quasi Peak/ 120 kHz	46	54	54	See b)
A12.3	30 – 300		46	54	50	See c)
	300 – 1 000	52				
A12.4	30 – 300	For frequencies ≥1 GHz	46	66	59	See d)
	300 – 1 000				52	
A12.5	30 – 950	Peak/ 1 MHz	46	76	46	See e)
	950 – 2 150			n/a	54	
a) Television receivers (analogue or digital), video recorders and PC TV broadcast receiver tuner cards working in channels between 30 MHz and 1 GHz, and digital audio receivers.						
b) Tuner units (not the LNB) for satellite signal reception.						
c) Frequency modulation audio receivers and PC tuner cards.						
d) Frequency modulation car radios.						
e) Applicable to EUTs with RF modulator output ports (for example DVD equipment, video recorders, camcorders and decoders etc.) designed to connect to TV broadcast receiver tuner ports.						

****Remarks: It is not necessary to be tested on this item.**

6. Differential Voltage Emissions

6.1 Test Setup and Procedure

6.1.1 Test Setup



6.1.2 Test Procedure

The output level of the auxiliary signal generator was set to 70dBuV at the EUT antenna terminal with 75 ohms impedance with an un-modulated carrier.

The highest emissions were analyzed in details by operating the spectrum analyzer and/or EMI receiver in quasi-peak mode to determine the precise amplitude of the emissions. The power of EUT was switched off to make sure the emission was not contributed by the auxiliary signal generator. While doing so, the interconnecting cables and major parts of the system were moved around to maximize the emission.

6.1.3 EMI Receiver Configuration (for the frequencies tested)

Frequency Range:	30MHz-2150MHz
Detector Function:	Quasi-Peak Mode
Resolution Bandwidth:	120kHz

6.1.4 Limit

Applicable to:

1. TV broadcast receiver tuner ports with an accessible connector.
2. RF modulator output ports.
3. FM broadcast receiver tuner ports with an accessible connector.

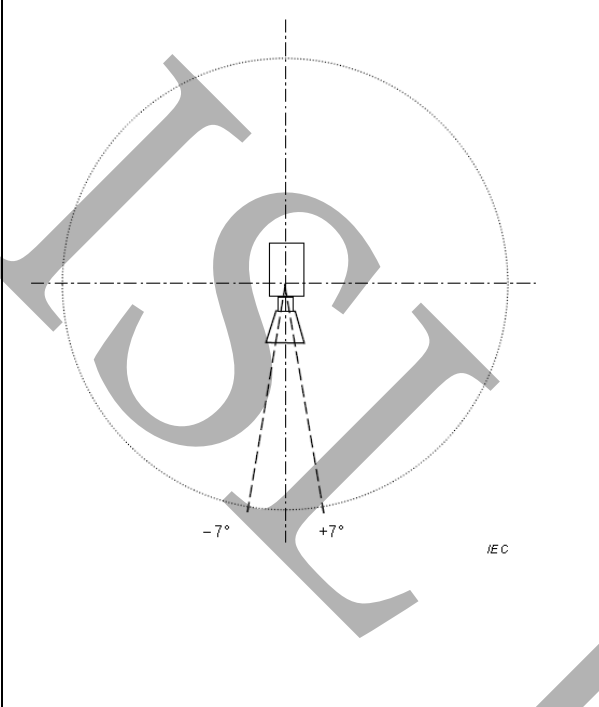
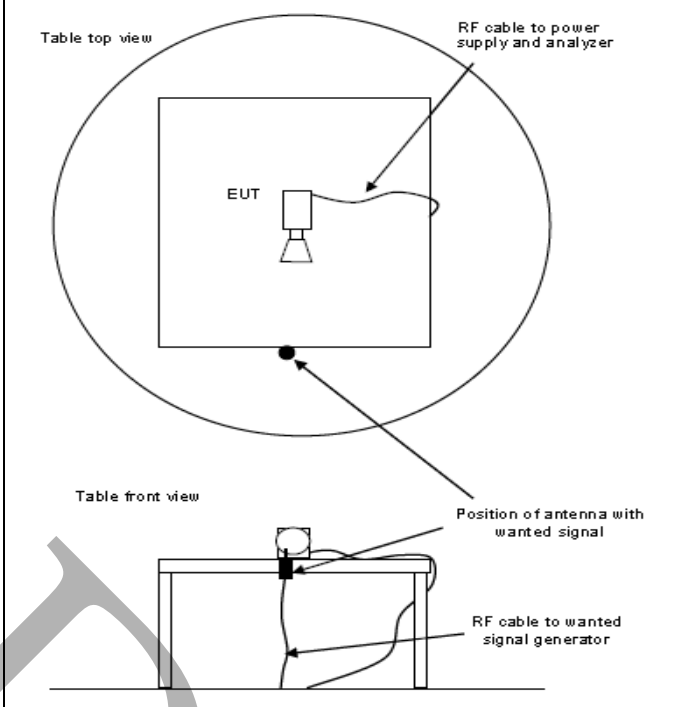
Table clause	Frequency range MHz	Detector type/ bandwidth	Class B limits dB(μV) 75 Ω			Applicability
			Other	Local Oscillator Fundamental	Local Oscillator Harmonics	
A12.1	30 – 950	For frequencies ≤1 GHz	46	46	46	See a)
	950 – 2 150		46	54	54	
A12.2	950 – 2 150	Quasi Peak/ 120 kHz	46	54	54	See b)
A12.3	30 – 300		46	54	50	See c)
	300 – 1 000	52				
A12.4	30 – 300	For frequencies ≥1 GHz	46	66	59	See d)
	300 – 1 000				52	
A12.5	30 – 950	Peak/ 1 MHz	46	76	46	See e)
	950 – 2 150			n/a	54	
a) Television receivers (analogue or digital), video recorders and PC TV broadcast receiver tuner cards working in channels between 30 MHz and 1 GHz, and digital audio receivers.						
b) Tuner units (not the LNB) for satellite signal reception.						
c) Frequency modulation audio receivers and PC tuner cards.						
d) Frequency modulation car radios.						
e) Applicable to EUTs with RF modulator output ports (for example DVD equipment, video recorders, camcorders and decoders etc.) designed to connect to TV broadcast receiver tuner ports.						

****Remarks: It is not necessary to be tested on this item.**

7. Outdoor units of home satellite receiving systems

7.1 Test Setup and Procedure

7.1.1 Test Setup

	
Description of $\pm 7^\circ$ of the main beam axis of the EUT	Measurement arrangements of transmit antenna for the wanted signal

7.1.2 Test Procedure

The input signal shall be adjusted to get the maximum rated output level from the EUT. For the measurement in the frequency range from 30 MHz to 18 GHz the input signal shall be adjusted so that the output frequency is within this frequency range. For the measurement in the frequency range above 1 GHz, the frequency of the input signal shall be adjusted in such a way that the EUT is measured, as a minimum, at the lowest, middle and highest rated output frequency within the measured frequency range.

7.1.3 Spectrum Analyzer Configuration (for the frequencies tested)

Frequency Range: 30MHz--1000MHz
 Detector Function: Quasi-Peak Mode
 Resolution Bandwidth: 120kHz

Frequency Range: Above 1000MHz
 Detector Function: Peak/Average Mode
 Resolution Bandwidth: 1MHz

7.1.4 Limit

Table Clause	Frequency Range MHz	Measurement			Class B Limits	Applicable to
		Facility (see Table A.1)	Distance m	Detector type / Bandwidth		
A7.1	30 to 1 000	SAC / OATS / FAR	See Table A.4	Quasi Peak / 120 kHz	See Table A.4	
A7.2	1 000 to 2 500	FSOATS	3	Average / 1 MHz	50 dB(μV/m)	LO leakage and spurious radiated emissions from the EUT, in the region outside ±7° of the main beam axis. See Figure H.1
	2 500 to 18 000				64 dB(μV/m)	
A7.3	1 000 to 18 000	FSOATS	3	Average / 1 MHz	37 dB(μV/m)	LO leakage from the EUT, in the region within ±7° of the main beam axis. See Figure H.1
A7.4	1 000 to 18 000	Conducted (Clause H.4)	n/a	Average / 1 MHz	30 dBpW	
For details of the EUT configuration, see Annex H.						
For radiated emissions measurements at frequencies up to 1 GHz, the requirements defined in Table A.4 shall be satisfied.						
Apply the appropriate limits across the entire frequency range.						
Apply the limits defined in table Clause A7.1 and A7.2. Also apply the limits defined in either table Clause A7.3 or A7.4.						

****Remarks: It is not necessary to be tested on this item.**

8. Electrostatic discharge (ESD) immunity

8.1 Test Specification and Setup

8.1.1 Test Specification

Port:	Enclosure
Basic Standard:	EN 61000-4-2/ IEC 61000-4-2 (details referred to Sec 1.2)
Test Level:	Air +/- 2 kV, +/- 4 kV, +/- 8 kV Contact +/- 4 kV
Criteria:	B
Test Procedure	refer to ISL QA -T4-E-S7
Temperature:	22°C
Humidity:	40%

Selected Test Point

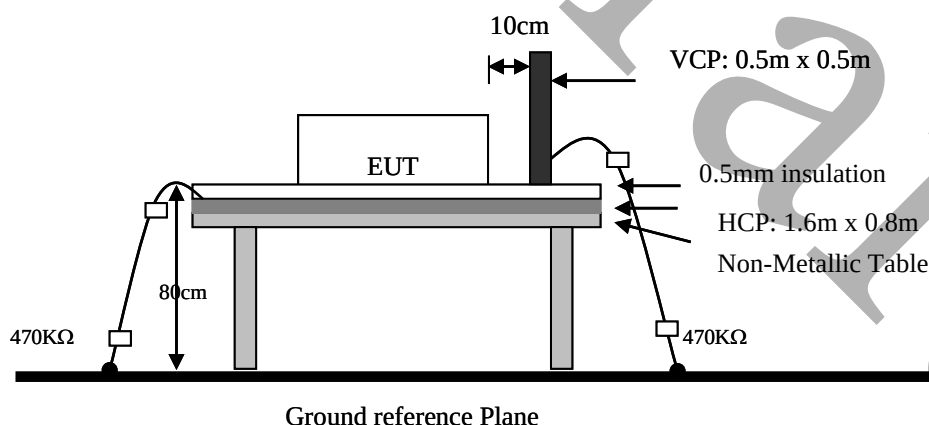
Air: discharges were applied to slots, aperture or insulating surfaces. 10 single air discharges were applied to each selected points.

Contact: 10 discharges minimum were to the selected contact points.

Indirect Contact Points: 10 discharges were applied to center of one edge of VCP and each EUT side of HCP with 10 cm away from EUT.

8.1.2 Test Setup

EUT is 1m from the wall and other metallic structure. When Battery test mode is needed, a cable with one 470K Ω resistor at two rare ends is connected from metallic part of EUT and screwed to HCP.



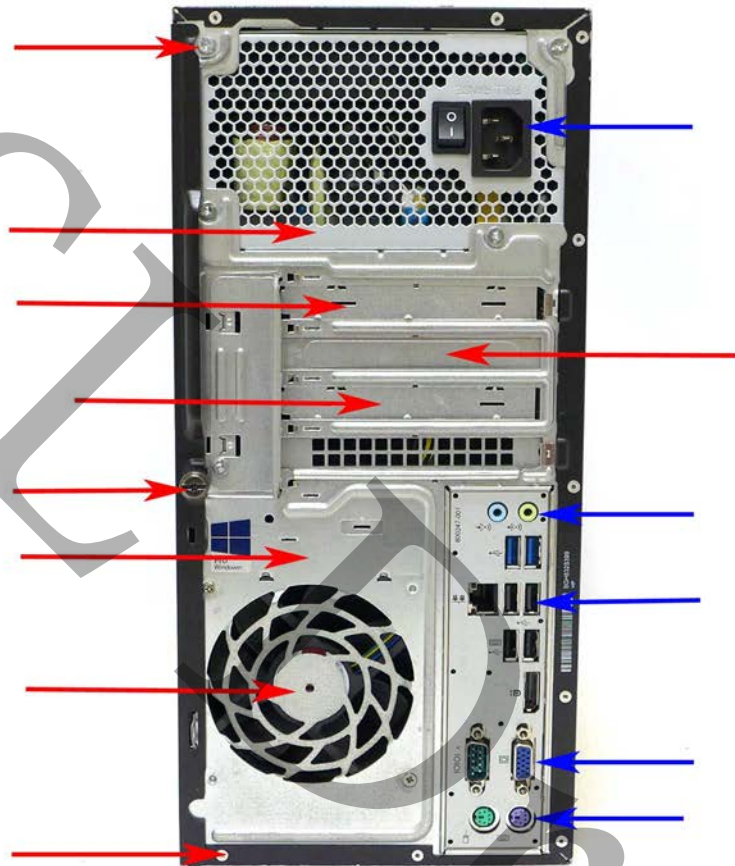
8.1.3 Test Result

Performance of EUT complies with the given specification

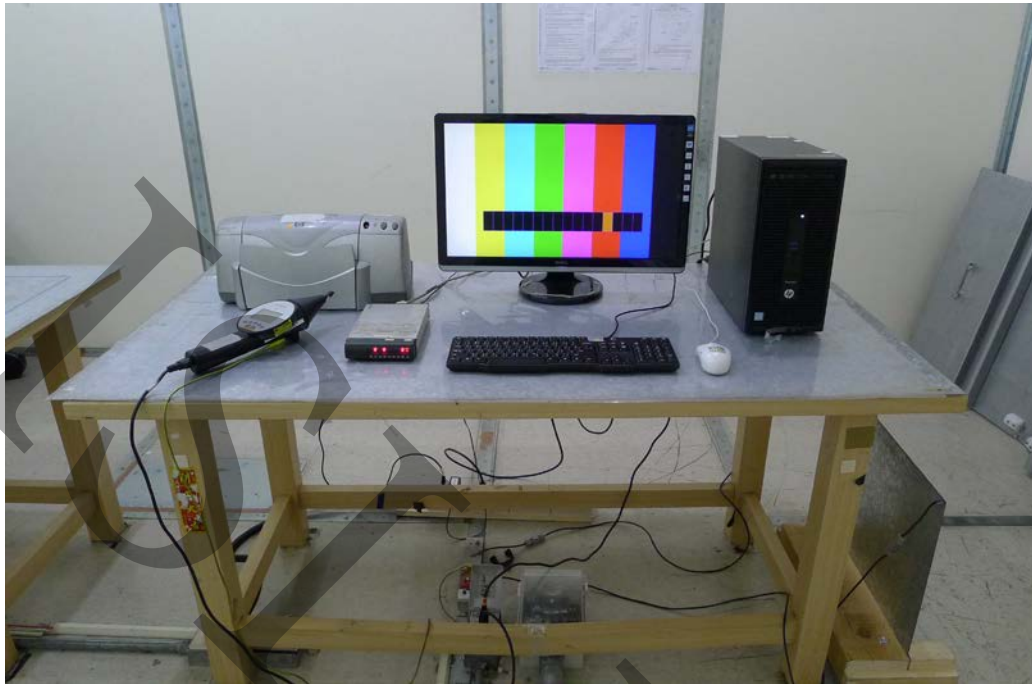
8.2 Test Point

Red arrow lines indicate the contact points, and blue arrow lines indicate the air points.

Figure 1: Test Point Assignments Discharge:



8.3 Test Setup Photo



9. Radio-Frequency, Electromagnetic Field immunity

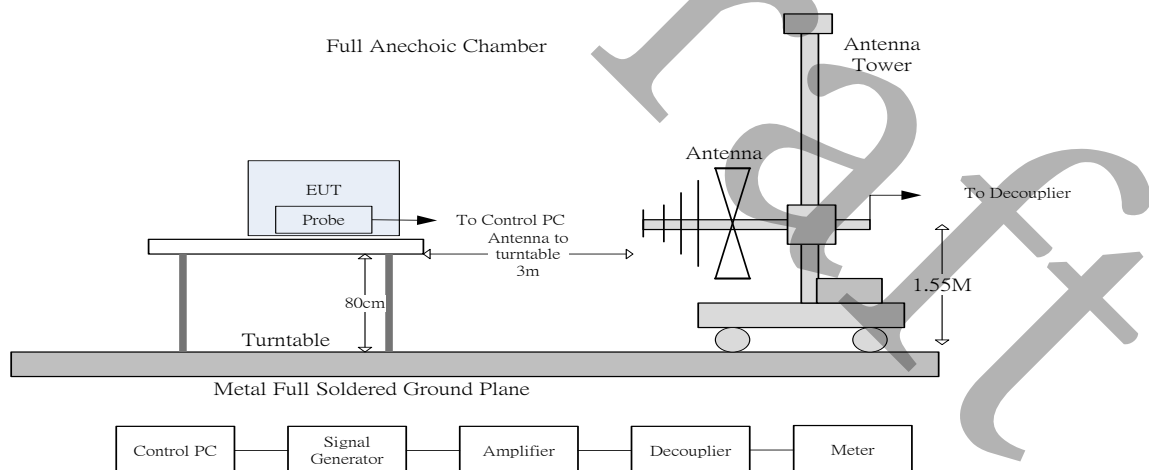
9.1 Test Specification and Setup

9.1.1 Test Specification

Port:	Enclosure
Basic Standard:	EN 61000-4-3/ IEC 61000-4-3 (details referred to Sec 1.2)
Test Level:	3 V/m
Modulation:	AM 1kHz 80%
Frequency range:	80 MHz~1 GHz 1800MHz, 2600MHz, 3500MHz, 5000MHz
Frequency Step:	1% of last step frequency
Dwell time:	3s
Polarization:	Vertical and Horizontal
EUT Azimuth Angle	☒ 0° ☒ 90° ☒ 180° ☒ 270°
Criteria:	A
Test Procedure	refer to ISL QA -T4-E-S8
Temperature:	24°C
Humidity:	55%

9.1.2 Test Setup

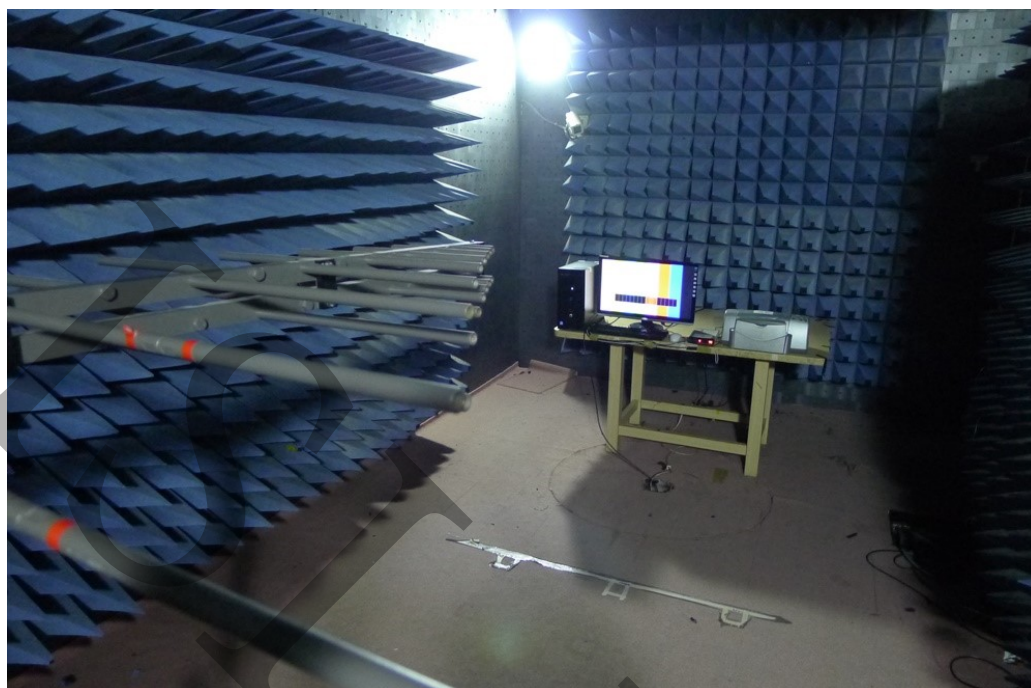
The field sensor is placed at one calibration grid point to check the intensity of the established fields on both polarizations. EUT is adjusted to have each side of EUT face coincident with the calibration plane. A CCD camera and speakers are used to monitor the condition of EUT for the performance judgment.



9.1.3 Test Result

Performance of EUT complies with the given specification

9.2 Test Setup Photo



10. Electrical Fast transients/burst immunity

10.1 Test Specification and Setup

10.1.1 Test Specification

Port:	AC mains
Basic Standard:	EN 61000-4-4/ IEC 61000-4-4 (details referred to Sec 1.2)
Test Level:	AC Power Port: +/- 1 kV
Rise Time:	5ns
Hold Time:	50ns
Burst Period:	300ms
Repetition Frequency:	5kHz
Criteria:	B
Test Procedure	refer to ISL QA -T4-E-S9
Temperature:	25 °C
Humidity:	55%

Test Procedure

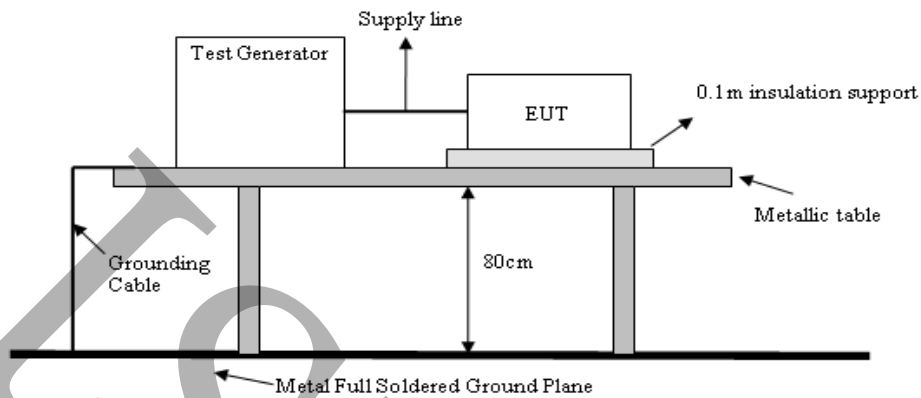
The EUT was setup on a nonconductive table 0.1 m above a reference ground plane.

Test Points	Polarity	Result	Comment
Line	+	N	60 sec
	-	N	60 sec
Neutral	+	N	60 sec
	-	N	60 sec
Ground	+	N	60 sec
	-	N	60 sec
Line to Neutral	+	N	60 sec
	-	N	60 sec
Line to Ground	+	N	60 sec
	-	N	60 sec
Neutral to Ground	+	N	60 sec
	-	N	60 sec
Line to Neutral to Ground	+	N	60 sec
	-	N	60 sec

Note: 'N' means normal, the EUT function is correct during the test.

10.1.2 Test Setup

EUT is at least 50cm from the conductive structure.



10.1.3 Test Result

Performance of EUT complies with the given specification

10.2 Test Setup Photo



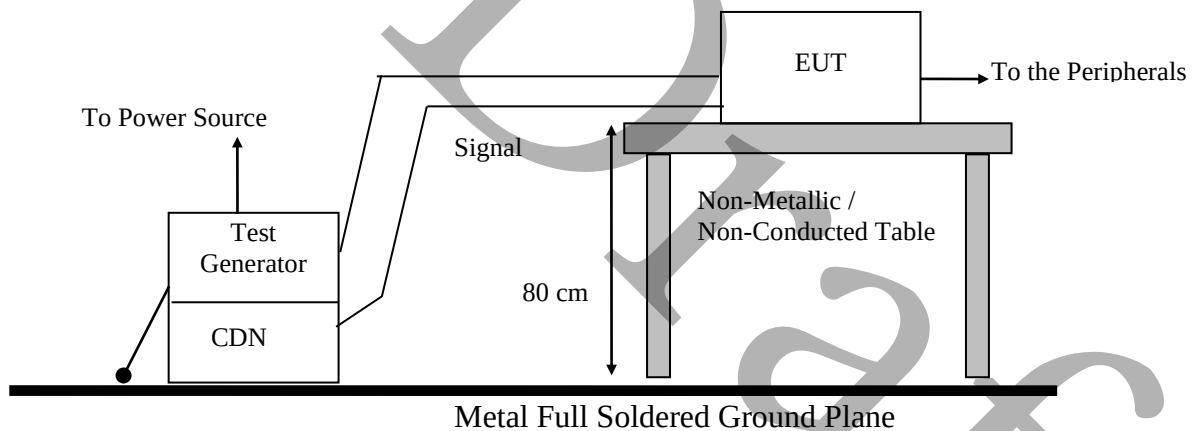
11. Surge Immunity

11.1 Test Specification and Setup

11.1.1 Test Specification

Port:	AC mains
Basic Standard:	EN 61000-4-5/ IEC 61000-4-5 (details referred to Sec 1.2)
Test Level:	Line to Line: +/- 0.5 kV, +/- 1 kV Line to Earth: +/- 0.5 kV, +/- 1 kV, +/- 2kV
Rise Time:	1.2us
Hold Time:	50us
Repetition Rate:	30 seconds
Angle:	☐ 0° ☒ 90° ☐ 180° ☒ 270°
Criteria:	B
Test Procedure:	refer to ISL QA -T4-E-S10
Temperature:	25°C
Humidity:	56%

11.1.2 Test Setup



11.1.3 Test Result

Performance of EUT complies with the given specification

11.2 Test Setup Photo



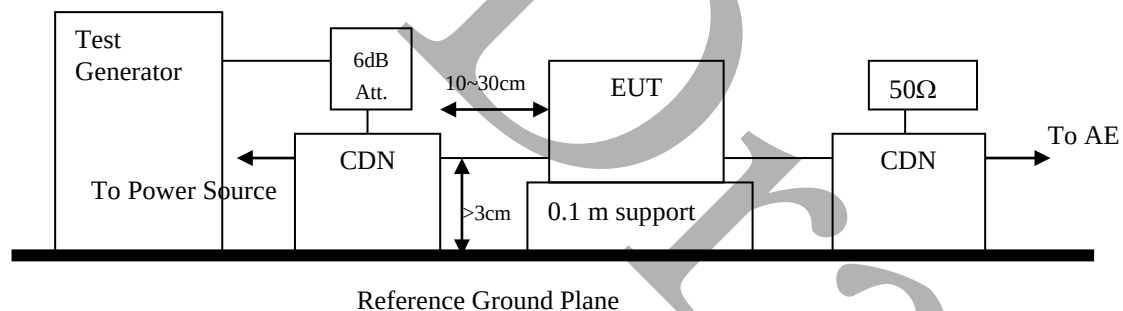
12. Immunity to Conductive Disturbance

12.1 Test Specification and Setup

12.1.1 Test Specification

Port:	AC mains
Basic Standard:	EN 61000-4-6/ IEC 61000-4-6 (details referred to Sec 1.2)
Frequency range and Test Level:	0.15MHz to 10MHz: 3 V 10MHz to 30MHz: 3V to 1V 30MHz to 80MHz: 1V
Modulation:	AM 1kHz 80%
Frequency Step:	1% of last Frequency
Dwell time:	3s
Criteria:	A
CDN Type:	CDN M2+M3
Test Procedure	refer to ISL QA -T4-E-S11
Temperature:	24°C
Humidity:	55%

12.1.2 Test Setup



12.1.3 Test Result

Performance of EUT complies with the given specification

12.2 Test Setup Photo



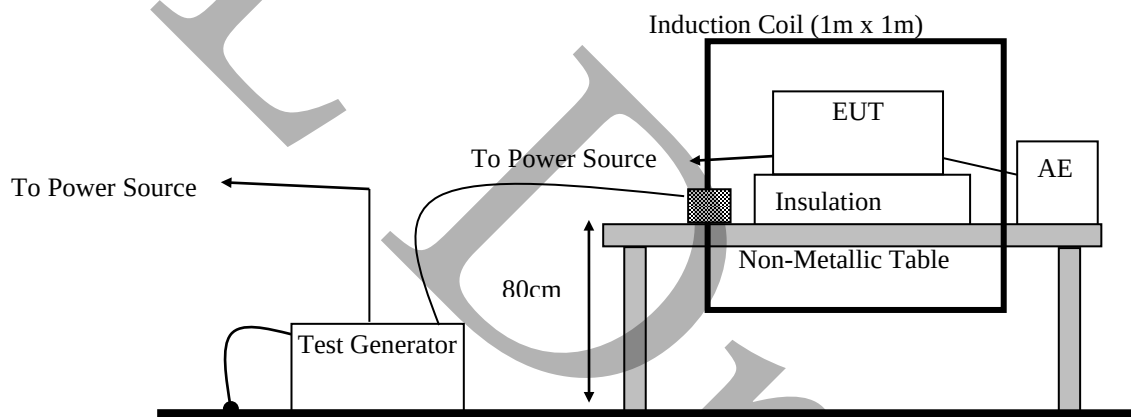
13. Power Frequency Magnetic Field immunity

13.1 Test Specification and Setup

13.1.1 Test Specification

Port:	Enclosure
Basic Standard:	EN 61000-4-8/ IEC 61000-4-8 (details referred to Sec 1.2)
Test Level:	1A/m
Polarization:	X, Y, Z
Criteria:	A
Test Procedure	refer to ISL QA -T4-E-S12
Temperature:	24°C
Humidity:	55%

13.1.2 Test Setup



13.1.3 Test Result

Performance of EUT complies with the given specification

13.2 Test Setup Photo



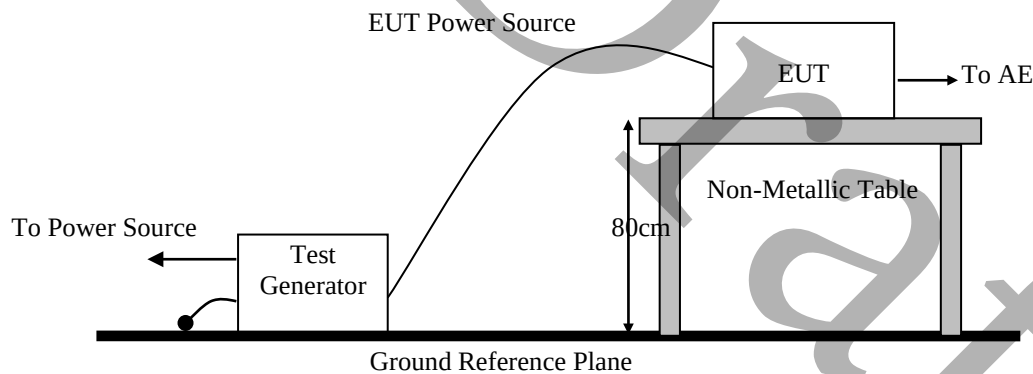
14. Voltage Dips, Short Interruption and Voltage Variation immunity

14.1 Test Specification and Setup

14.1.1 Test Specification

Port:	AC mains
Basic Standard:	EN 61000-4-11/ IEC 61000-4-11 (details referred to Sec 1.2)
Test Level: Criteria:	>95% in 0.5 cycle B
Test Level: Criteria:	30% in 25 cycle C
Test Level: Criteria:	>95% in 250 cycle C
Phase:	0°; 180°
Test intervals:	3 times with 10s each
Test Procedure	refer to ISL QA -T4-E-S13
Temperature:	25°C
Humidity:	56%

14.1.2 Test Setup



14.1.3 Test Result

Performance of EUT complies with the given specification

14.2 Test Setup Photo



15. Harmonics

15.1 Test Specification and Setup

15.1.1 Test Specification

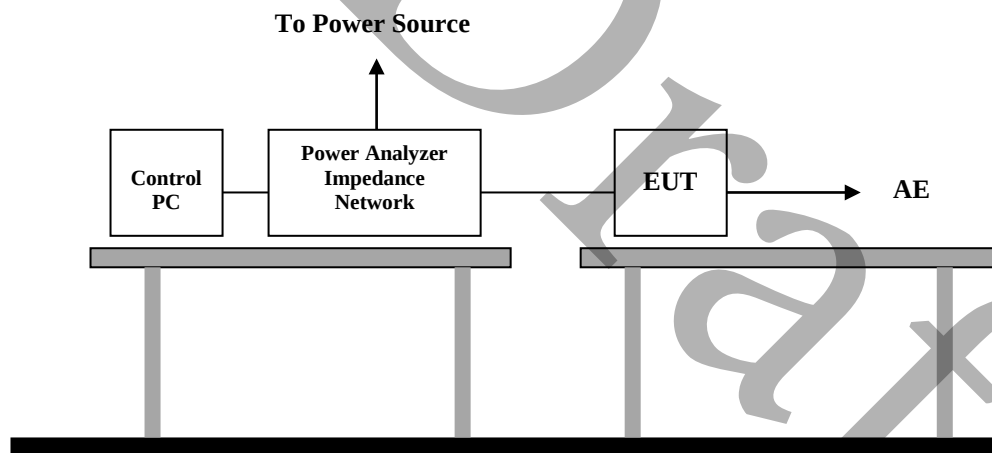
Port:	AC mains
Active Input Power:	<75W
Basic Standard:	EN 61000-3-2/IEC 61000-3-2 (details referred to Sec 1.2)
Test Duration:	2.5min
Class:	A
Test Procedure	refer to ISL QA -T4-E-S14
Temperature:	24°C
Humidity:	55%

Test Procedure

The EUT is supplied in series with shunts or current transformers from a source having the same nominal voltage and frequency as the rated supply voltage and frequency of the EUT. The EUT is configured to its rated current with additional resistive load when the testing is performed.

Equipment having more than one rated voltage shall be tested at the rated voltage producing the highest harmonics as compared with the limits.

15.1.2 Test Setup



15.1.3 Limit

Limits of Class A Harmonics Currents

Harmonics Order n	Maximum Permissible harmonic current A	Harmonics Order n	Maximum Permissible harmonic current A
Odd harmonics		Even harmonics	
3	2.30	2	1.08
5	1.14	4	0.43
7	0.77	6	0.30
9	0.40	$8 \leq n \leq 40$	$0.23 * 8/n$
11	0.33		
13	0.21		
$15 \leq n \leq 39$	$0.15 * 15/n$		

15.1.4 Test Result

Active input power under 75W, no limit apply, declare compliance

16. Voltage Fluctuations

16.1 Test Specification and Setup

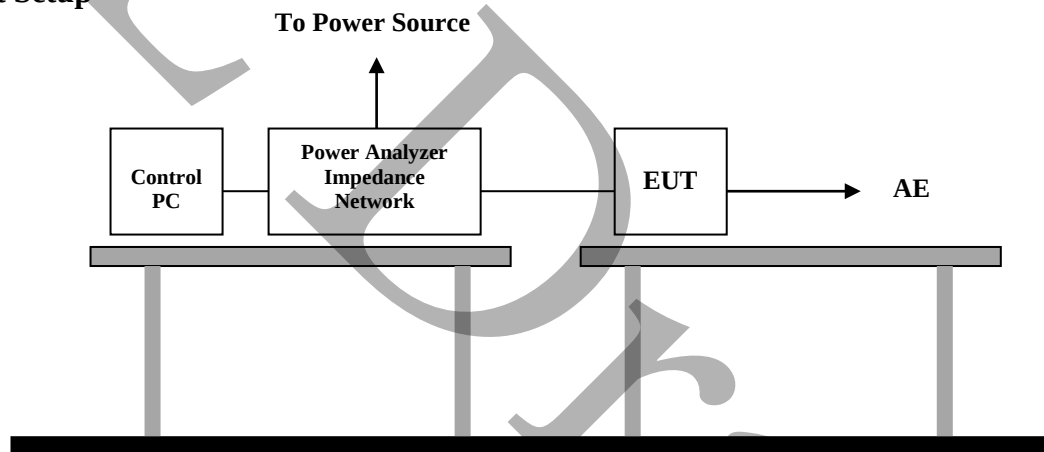
16.1.1 Test Specification

Port:	AC mains
Basic Standard:	EN 61000-3-3/IEC 61000-3-3 (details referred to Sec 1.2)
Test Procedure	refer to ISL QA -T4-E-S14
Observation period:	For Pst 10min
	For Plt 2 hours
Temperature:	24°C
Humidity:	55%

Test Procedure

The EUT is supplied in series with reference impedance from a power source with the voltage and frequency as the nominal supply voltage and frequency of the EUT.

16.1.2 Test Setup



16.1.3 Test Result

Performance of EUT complies with the given specification.

16.2 Test Data

Flicker Test Summary per EN/IEC61000-3-3 Ed. 3.0 (2013) (Run time)

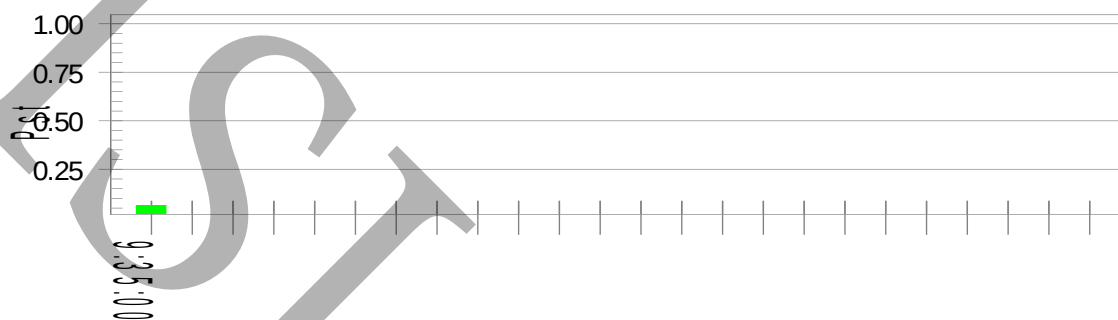
Test category: All parameters (European limits) Test Margin: 100
Test duration (min): 10 Data file name: CTSMXL_F-000303.cts_data

Test Result: Pass

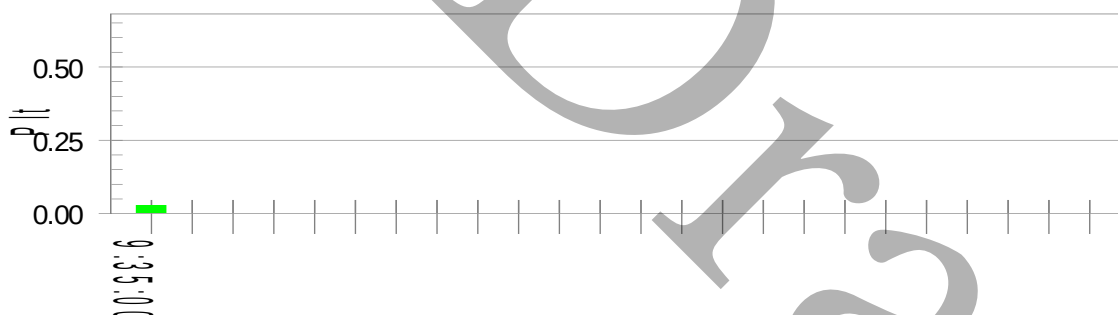
Status: Test Completed

Pst_i and limit line

European Limits



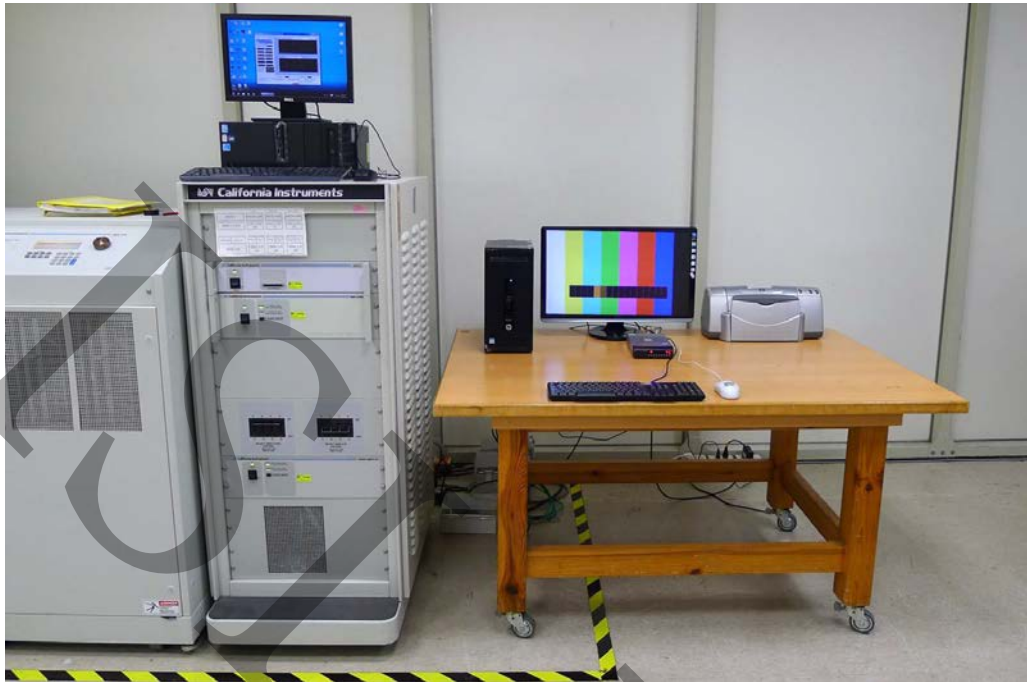
Plt and limit line



Parameter values recorded during the test:

Vrms at the end of test (Volt):	229.62	Test limit (mS):	500.0	Pass
T-max (mS):	0.0	Test limit (%):	3.30	Pass
Highest dc (%):	0.00	Test limit (%):	4.00	Pass
Highest dmax (%):	0.07	Test limit:	1.000	Pass
Highest Pst (10 min. period):	0.064	Test limit:	0.650	Pass
Highest Plt (2 hr. period):	0.029			

16.3 Test Setup Photo



17. Appendix

17.1 Appendix A: Test Equipment

17.1.1 Test Equipment List

Location Con03	Equipment Name	Brand	Model	S/N	Last Cal. Date	Next Cal. Date
Conduction 03	EMI Receiver 13	ROHDE & SCHWARZ	ESCI	101015	07/25/2019	07/25/2020
Conduction 03	LISN 15	R&S	ENV216	101335	11/22/2018	11/22/2019
Conduction 03	LISN 08	FCC	FCC-LISN-50/250-25-2-01	07039	08/14/2019	08/14/2020
Conduction 03	Conduction 03 -1 Cable	WOKEN	CFD 300-NL	Conduction 03 -1	08/30/2018	08/30/2019

Location Chamber02	Equipment Name	Brand	Model	S/N	Last Cal. Date	Next Cal. Date
Radiation (Chamber02)	BILOG Antenna 17	Schwarzbeck	Schwarzbeck VULB 9168+EMCI-N-6-05	645	03/06/2019	03/06/2020
Radiation (Chamber02)	Preamplifier 25	EMCI	EMC9135	980295	02/27/2019	02/27/2020
Radiation (Chamber02)	Coaxial Cable Chmb 02-10M-02	EMC	RG214U	Chmb 02-10M-02	08/30/2018	08/30/2019
Radiation (Chamber02)	EMI Receiver 12	ROHDE & SCHWARZ	ESCI	100804	08/14/2019	08/14/2020

Location Chmb14	Equipment Name	Brand	Model	S/N	Last Cal. Date	Next Cal. Date
Rad. Above 1GHz	Spectrum Analyzer 25	R&S	FSV 40	101499	11/03/2018	11/03/2019
Rad. Above 1GHz	Horn Antenna 06	ETS-Lindgren	3117	00066665	10/31/2018	10/31/2019
Rad. Above 1GHz	Preamplifier 13	MITEQ	JS44-00101800-25-10P-44	1329256	11/21/2018	11/21/2019
Rad. Above 1GHz	Microwave Cable 35	WOKEN	WCBA-WCA04NM.SM6	Chamber 14-1	01/31/2019	01/31/2020
Rad. Above 1GHz	Microwave Cable 36	WOKEN	WCBA-WCA04NM.SM0.8	Chamber 14-2	01/31/2019	01/31/2020

Location	Equipment Name	Brand	Model	S/N	Last Cal. Date	Next Cal. Date
EN61K-4-2	ESD Gun 12	EM TEST	Dito	P1650188689	05/07/2019	05/07/2020
EN61K-4-3	Broadband Log-Periodic Antenna	AR	AT1080	310698	N/A	N/A
EN61K-4-3	Horn Antenna RF-01	AR	ATS700M11 G	0335864	N/A	N/A
EN61K-4-3	Amplifier 80Mz~1GHz 250W	AR	250W1000A	312494	N/A	N/A
EN61K-4-3	Amplifier 800MHz~4.2GHz 50W	AR	50S1G4M1	312762	N/A	N/A
EN61K-4-3	Amplifier 4.0~8.0GHz 35W	AR	35S4G8AM1	0335752	N/A	N/A
EN61K-4-3	Broadband Coupler 80M~1GHz	Amplifier Research	DC6180A	0341805	N/A	N/A
EN61K-4-3	Coaxial Cable	INSULATED	NPS-4806-23 60-NP3	108599.003.01.03	N/A	N/A
EN61K-4-3	Broadband Coupler 0.8G~4.26GHz	AR	DC7144A	0335226	N/A	N/A
EN61K-4-3	Broadband Coupler 4G~8GHz	AR	DC7350A	0335817	N/A	N/A
EN61K-4-3	Signal Generator 07	ROHDE& SCHWARZ	SMB100A	107780	10/28/2018	10/28/2019
EN61K-4-4	EFT and SURGE Test System	EM TEST	UCS-500 M6B	V0728102674	02/14/2019	02/14/2020
EN61K-4-5	CDN-UTP8 ED3	EMC-PARTNER	CDN-UTP8	1509	04/02/2019	04/02/2020
EN61K-4-5	SURGE-TESTER	EMC Partner	MIG0603IN3	523	04/02/2019	04/02/2020
EN61K-4-6	CDN M2+M3 05	Frankonia	CDN M2+M3	A2210235/2013	09/22/2018	09/22/2019
EN61K-4-6	Coaxial Cable 4-6 02-1			4-6 02-1	N/A	N/A
EN61K-4-6	Conducted Immunity Test System 02	Frankonia	CIT-10-75-D C	126B1301/2014	03/25/2019	03/25/2020
EN61K-4-8	Magnetic Field Immunity Loop	FCC	F-1000-4-8-L-1M	01037	05/27/2019	06/05/2020
EN61K-4-8	Magnetic Field Test Generator	FCC	F-1000-4-8-G-125A	01038	05/27/2019	06/05/2020
EN61K-4-11	Voltage Dip and UP Simulator	NoiseKen	VDS-2002	VDS0640162	11/06/2018	11/06/2019
EN61K-3-2/3, EN61K-3-11-1 2	(Harmonic/Flicker) MX Series CTS Compliance Test System	California Instruments	MX60T04GH 10400	72793	08/05/2019	08/05/2020

PS: N/A => The equipment does not need calibration.

****Software for Controlling Spectrum/Receiver and Calculating Test Data**

Test Item	Filename	Version
EN61000-3-2	California Instruments	CTSMXL V2.19.0
EN61000-3-3	California Instruments	CTSMXL V2.19.0
EN61000-4-2	N/A	2.0
EN61000-4-3	i2	4.130102k
EN61000-4-4	EMC TEST	4.10
EN61000-4-5	EMC Partner	1.69
EN61000-4-6	FRANKONIA CD-LAB	V5.221
EN61000-4-8	N/A	
EN61000-4-11	NOISE KEN	2.0

Site	Filename	Version	Issue Date
Conduction/Radiation	EZ EMC	ISL-03A2	3/6/2013

17.2 Appendix B: Uncertainty of Measurement

The measurement uncertainty refers to CISPR 16-4-2. The coverage factor $k = 2$ yields approximately a 95 % level of confidence.

<Conduction 03>

AMN: $\pm 2.90\text{dB}$

<Chamber 02 (10M)>

Horizontal

30MHz~200MHz: $\pm 4.69\text{dB}$

200MHz~1000MHz: $\pm 4.30\text{dB}$

Vertical

30MHz~200MHz: $\pm 4.65\text{dB}$

200MHz~1000MHz: $\pm 4.35\text{dB}$

<Chamber 14 (3M)>

1GHz~6GHz: $\pm 5.12\text{dB}$

<Immunity 02>

Test item	Uncertainty	Test item	Uncertainty
EN 61000-4-2 (ESD)		EN 61000-4-6 (CS)	
Rise time tr	$\leq 15\%$	CDN	$\pm 1.36\text{dB}$
Peak current Ip	$\leq 6.3\%$	EM Clamp	$\pm 3.19\text{dB}$
current at 30 ns	$\leq 6.3\%$	EN 61000-4-8 (Magnetic)	$\pm 6.55\%$
current at 60 ns	$\leq 6.3\%$	EN 61000-4-11 (Dips)	
EN 61000-4-3 (RS)	$\pm 2.19\text{dB}$	Time	$\pm 2.80\%$
EN 61000-4-4 (EFT)		Voltage	$\pm 1.5\%$
voltage rise time (tr)	$\pm 6.2\%$	EN 61000-3-2 (Harmonics)	$\pm 5.1 \%$
peak voltage value (VP)	$\pm 8.6\%$	EN 61000-3-3 (Fluctuations and Flicker)	$\pm 5.1 \%$
voltage pulse width (tw)	$\pm 5.9\%$		
EN 61000-4-5 (Surge)			
open-circuit voltage front time	$\pm 1.2\mu\text{s}$		
open-circuit voltage peak value	$\pm 8.6\%$		
open-circuit voltage duration (Td)	$\pm 50.7\mu\text{s}$		
EN 61000-4-9 (Pulse magnetic field)			
Time	$\pm 2.80\%$		
Voltage	$\pm 5.5\%$		
Current	$\pm 3.3\%$		

17.3 Appendix C: Photographs of EUT

Please refer to the File of **ISL-19LE455P-MB**

ISL Draft